

Perception & Color

COMP7507
Visualization & Visual Analytics



SCHOOL OF
**COMPUTING &
DATA SCIENCE**
The University of Hong Kong

What is Perception?

- **Perception** is the process by which we interpret the world around us, forming a mental representation of the environment.
- Processing of our sensory information
 - **Recognizing** (being aware of, more like detection)
 - **Organizing** (gathering and storing)
 - **Interpreting** (binding to knowledge)
- Subject to **correspondence differences and errors**

Further reading (for interest):

Visual Perception Theory (<https://www.simplypsychology.org/perception-theories.html>)

Why is something easier to find?

- Find the p's
- Find the q's

ehklhfdiyaoryweklblkhockxlyhirhupw
lksajdhflkihqdaklljerlajesljselusdsfjsals
oufojrto

pjhklghqlkshlkfhlkdshflymcvci

- What makes an object easy to find is how easily we can [direct a rapid eye movement](#) to focus our attention on it.

Eye Movement and Attention

- When we are looking at something, we **focus** & **process**.

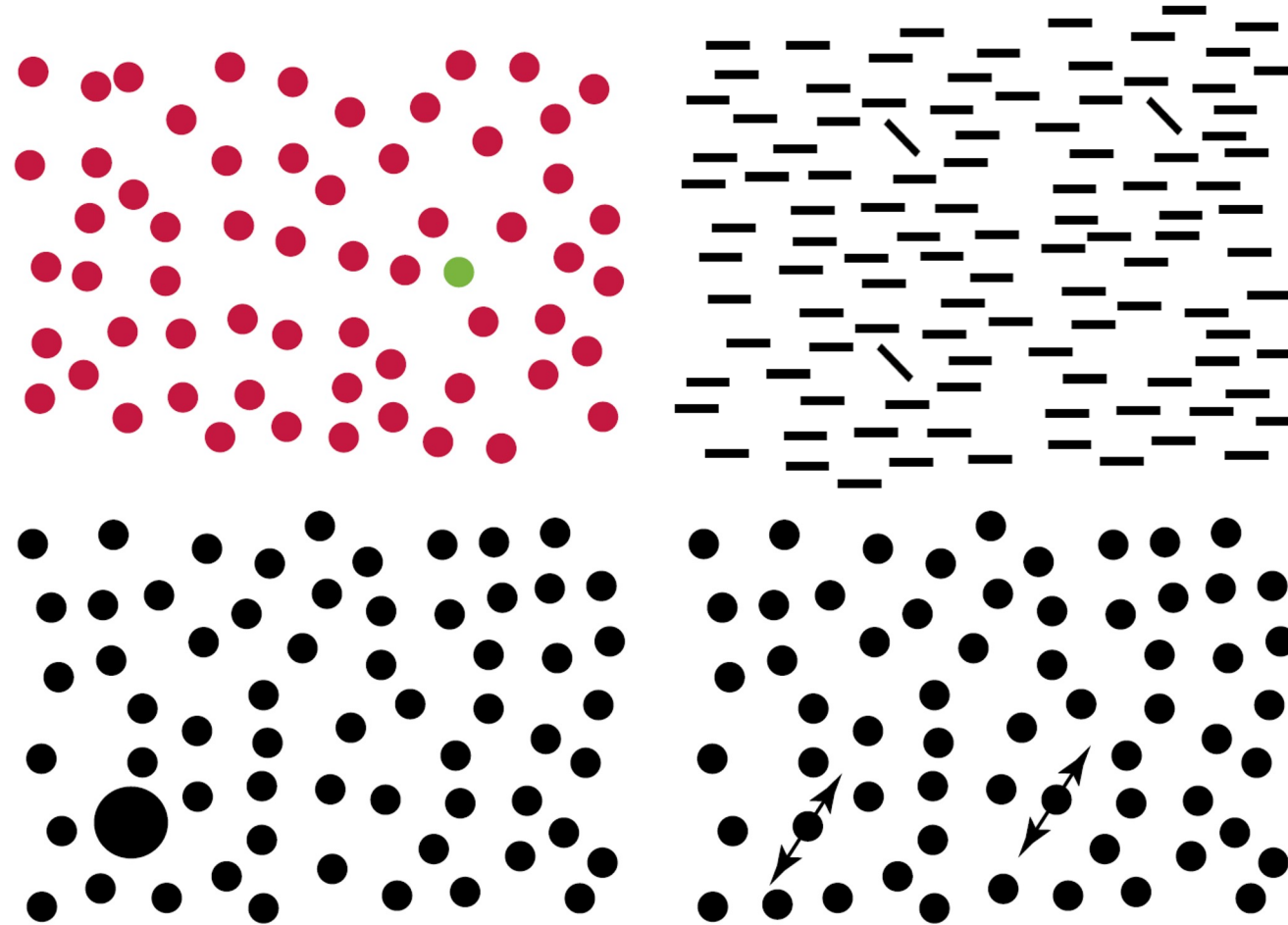


Find the tomatoes

Eye Movement and Attention

- How to get our eyes focused at something in the absence of prior knowledge of where things are?
 - Important for **exploratory visualization**
- How to draw one's attention? What will stand out in a graphic?

Eye Movement Planning



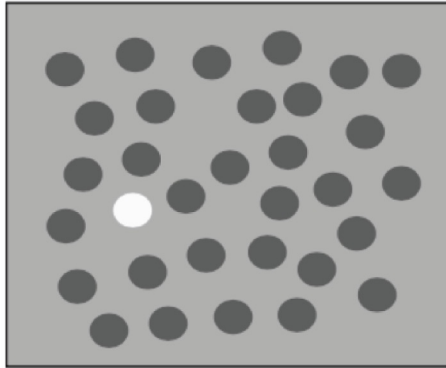
[Colin
2008]

Some features has a **pop-out effect** that we can automatically program an eye movement and focus our attention there.

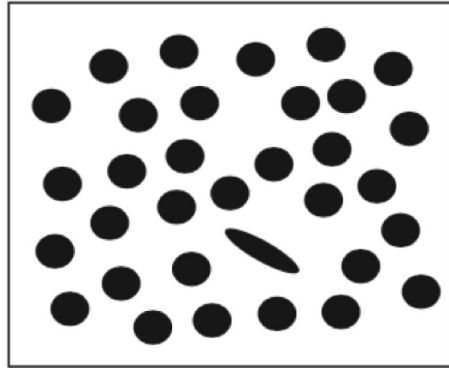
Preattentive Processing

- The **automatic** mechanisms that operate **prior to** the action of attention
- **Uncontrolled** perception
- The **fast** ability to detect features, usually at a rate faster than 10 msec per item
- Time taken to find the target is independent of the number of distracting nontargets
- Allows detecting features in **parallel**

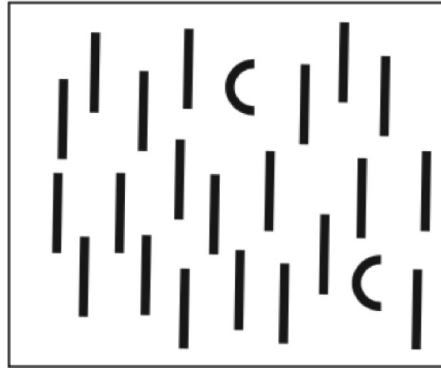
Preattentive Features



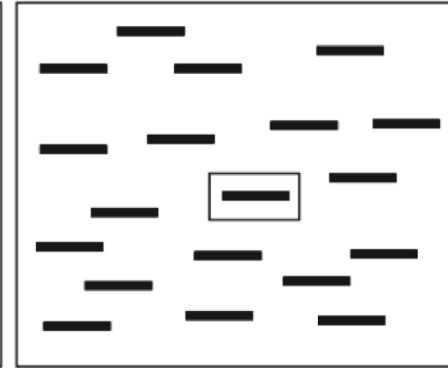
Grey value



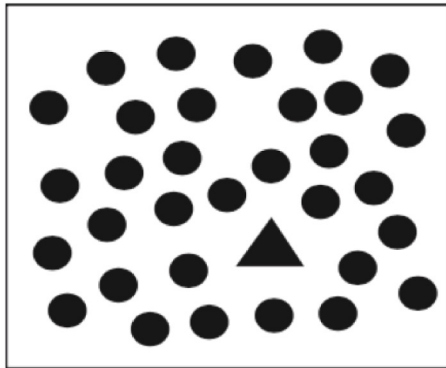
Elongation



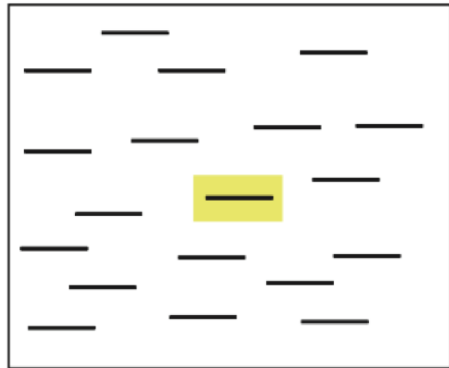
Curvature



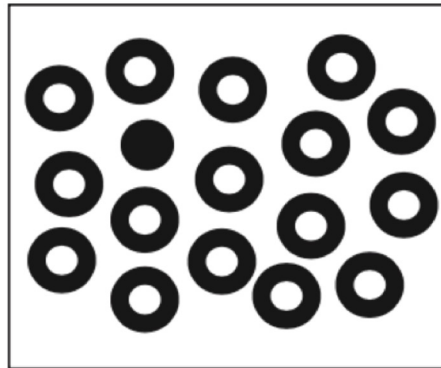
Added surround box



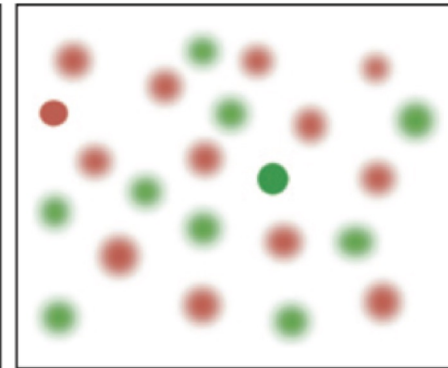
Shape



Added surround color

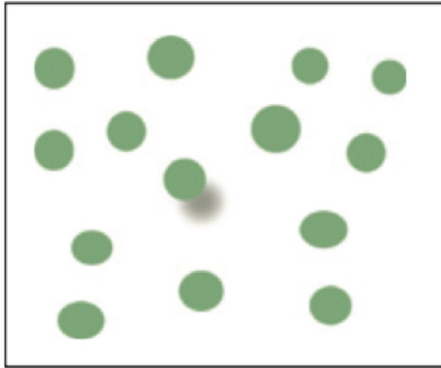


Filled

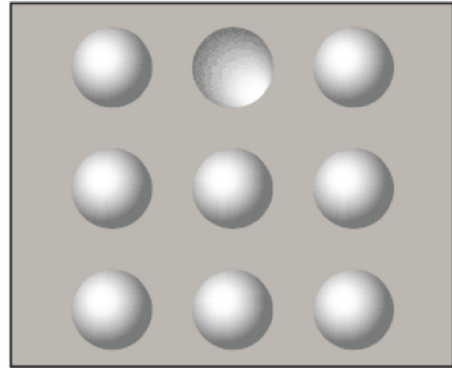


Sharpness

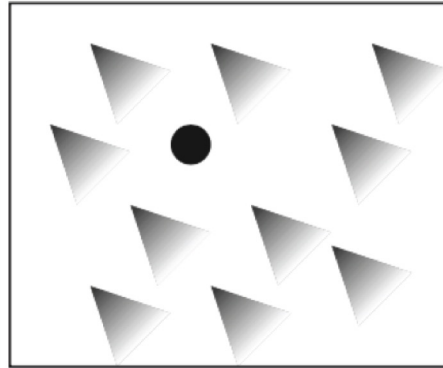
Preattentive Features



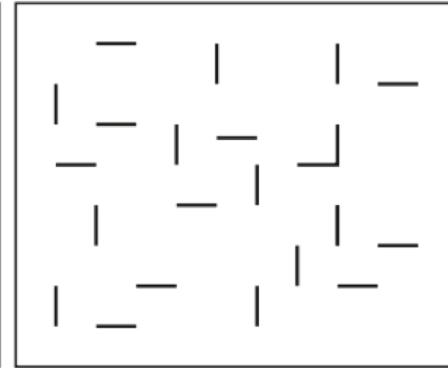
Shadow



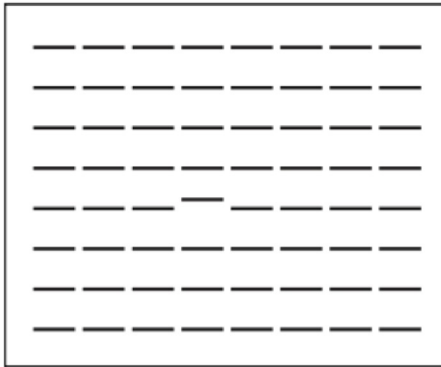
Convex and concave



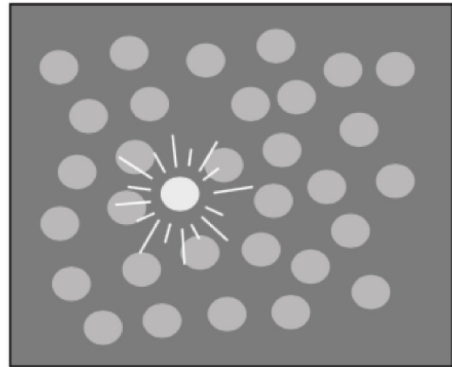
Sharp vertex



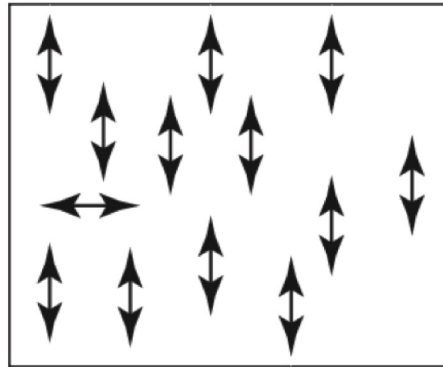
Joined lines



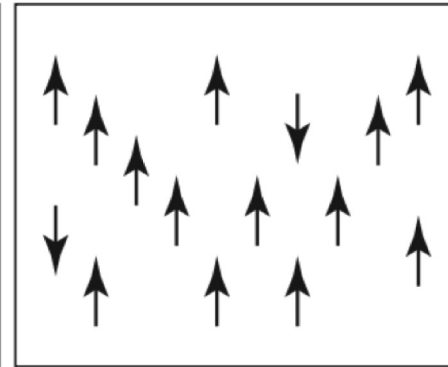
Alignment



Blinking



Motion direction

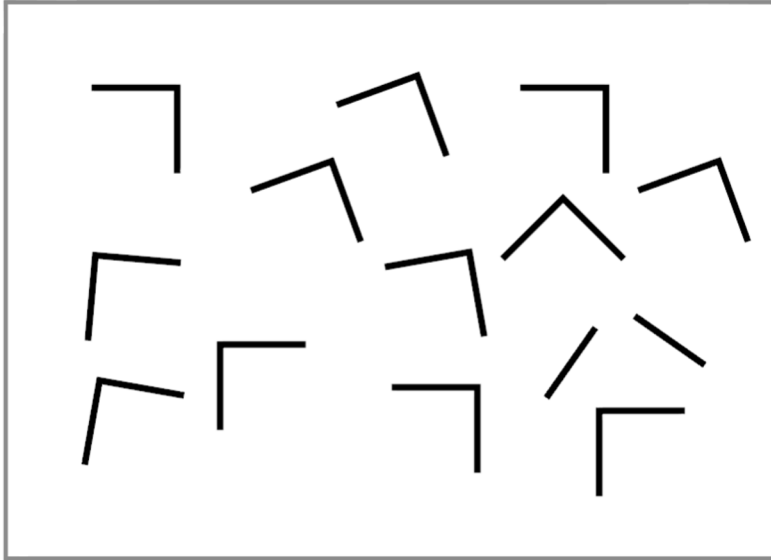


Phase of motion

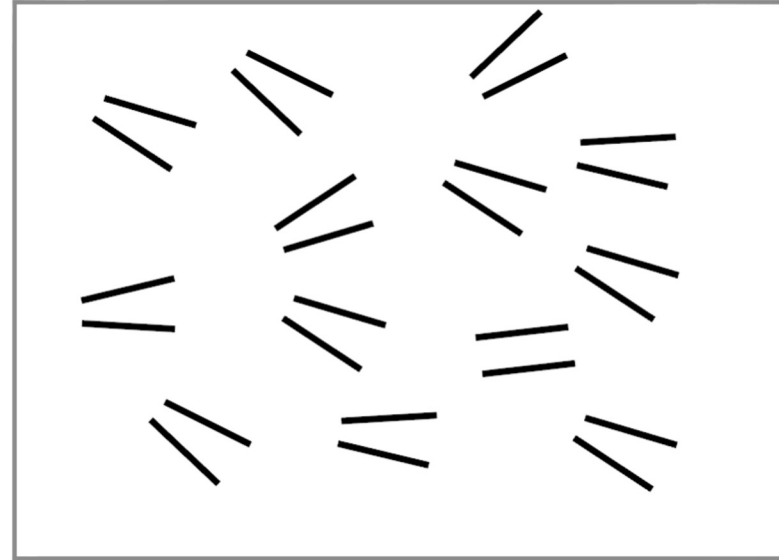
[Colin 2008]

Non-Preattentive Features

- Not all features are preattentive



Juncture



Parallelism

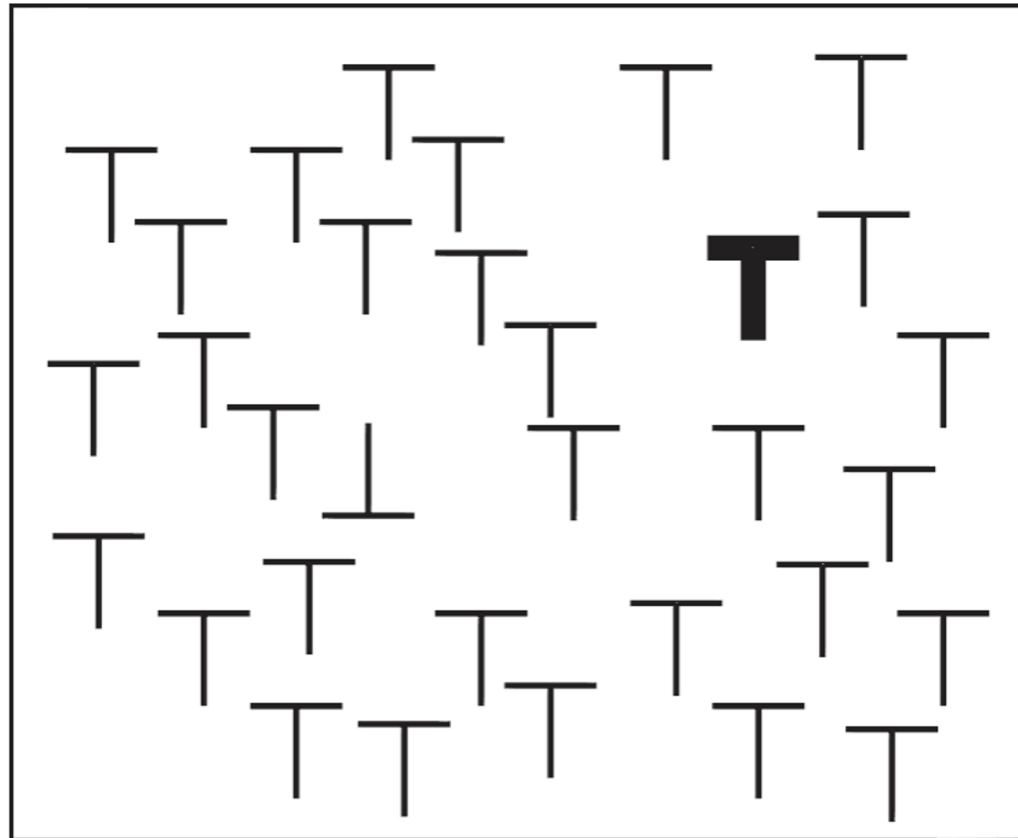
[Colin 2008]

Easy vs. Hard to Find

- Some features are easier to find:

Find 

Find 



[Colin 2008]

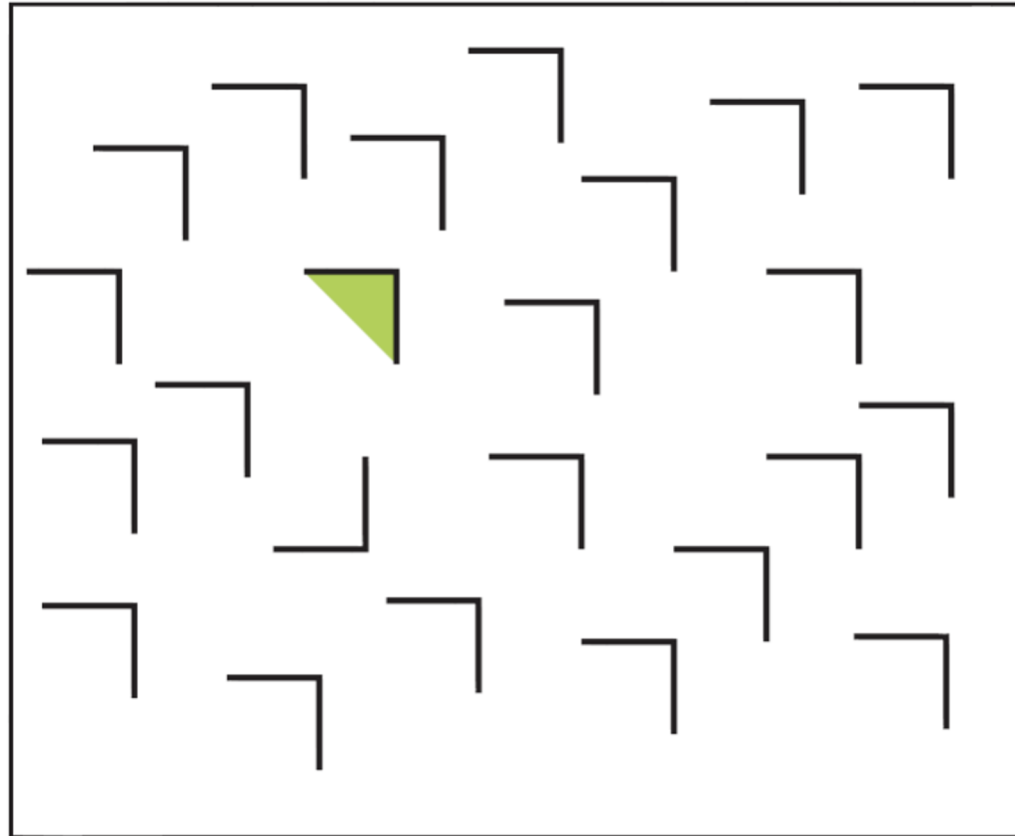
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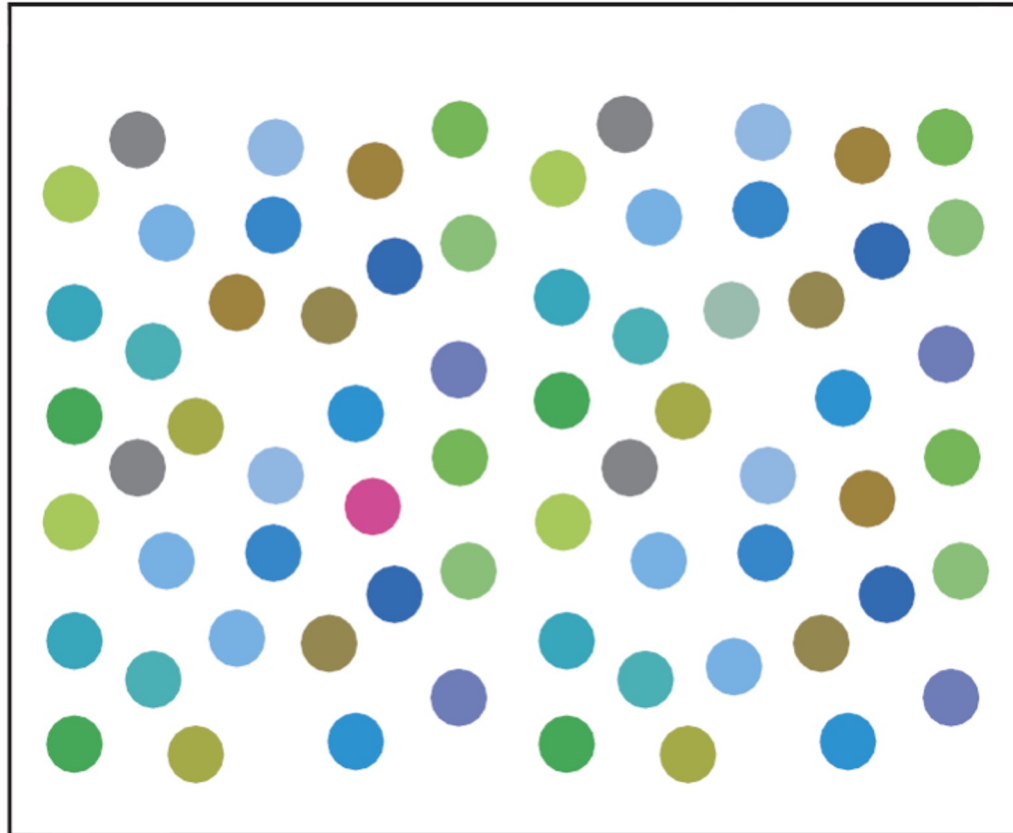


Easy vs. Hard to Find

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Find 



[Colin 2008]

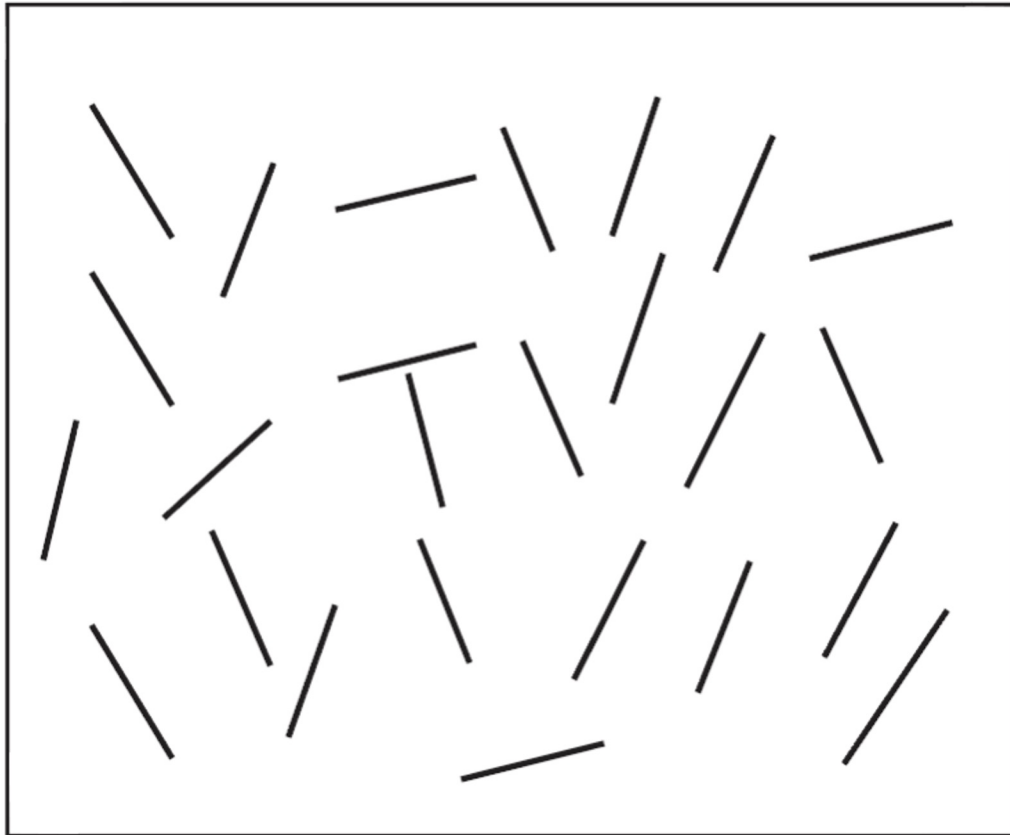
Easy vs. Hard to Find

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Find

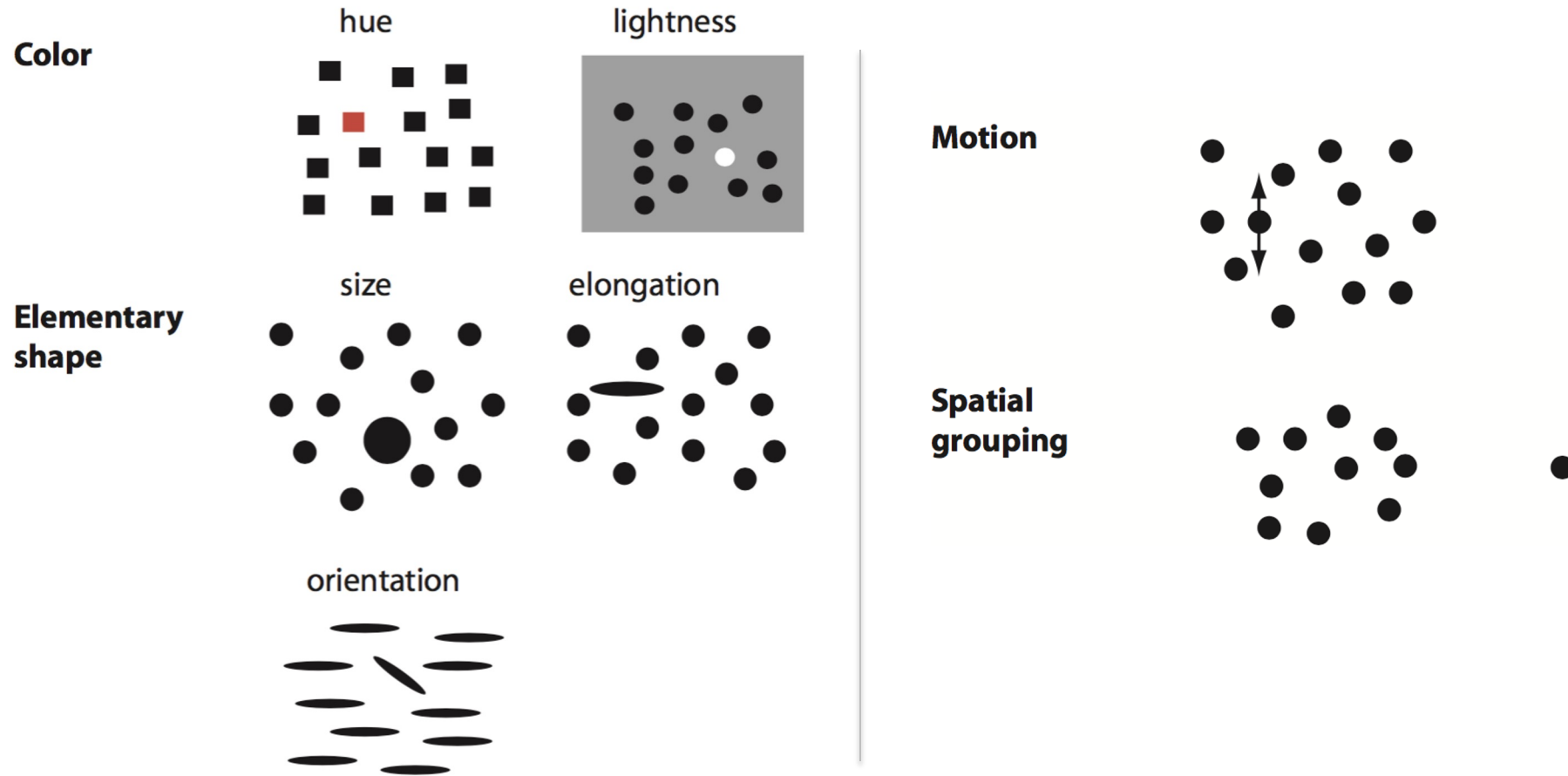


Find



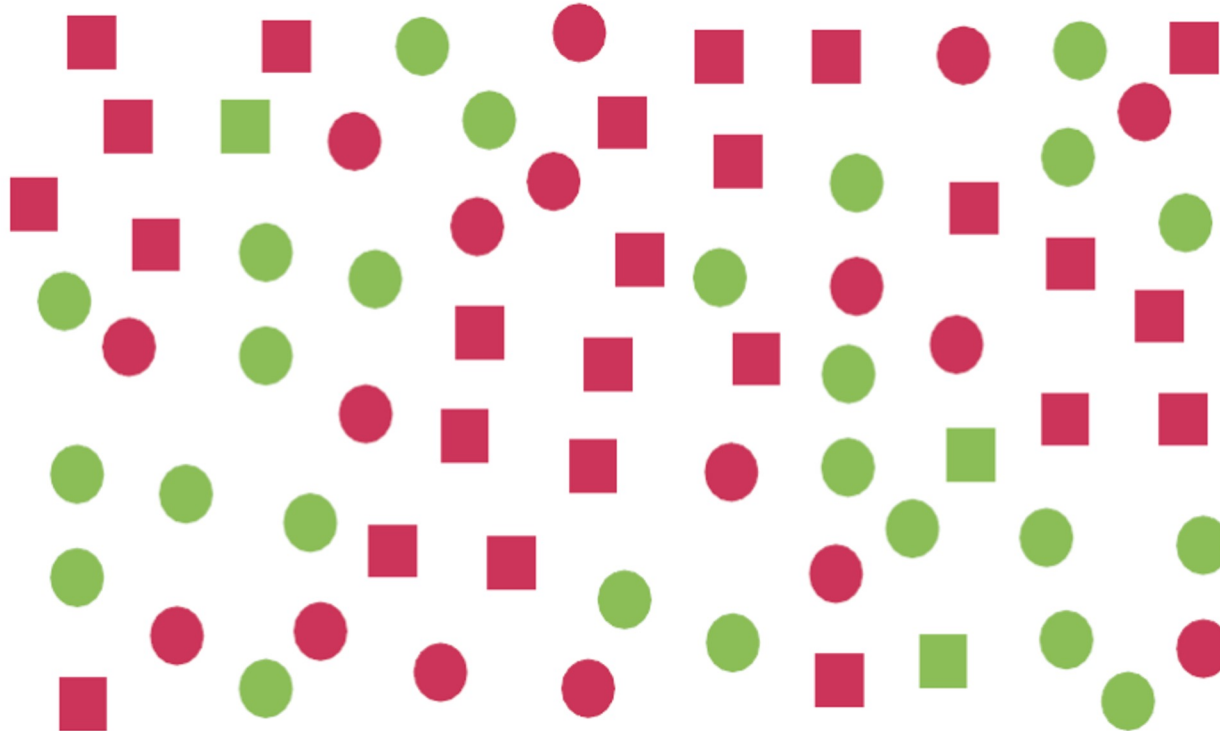
[Colin 2008]

Basic Popout Channels



Visual Conjunctions

- conjunctive search
 - to find a target with two features
 - generally NOT pre-attentive

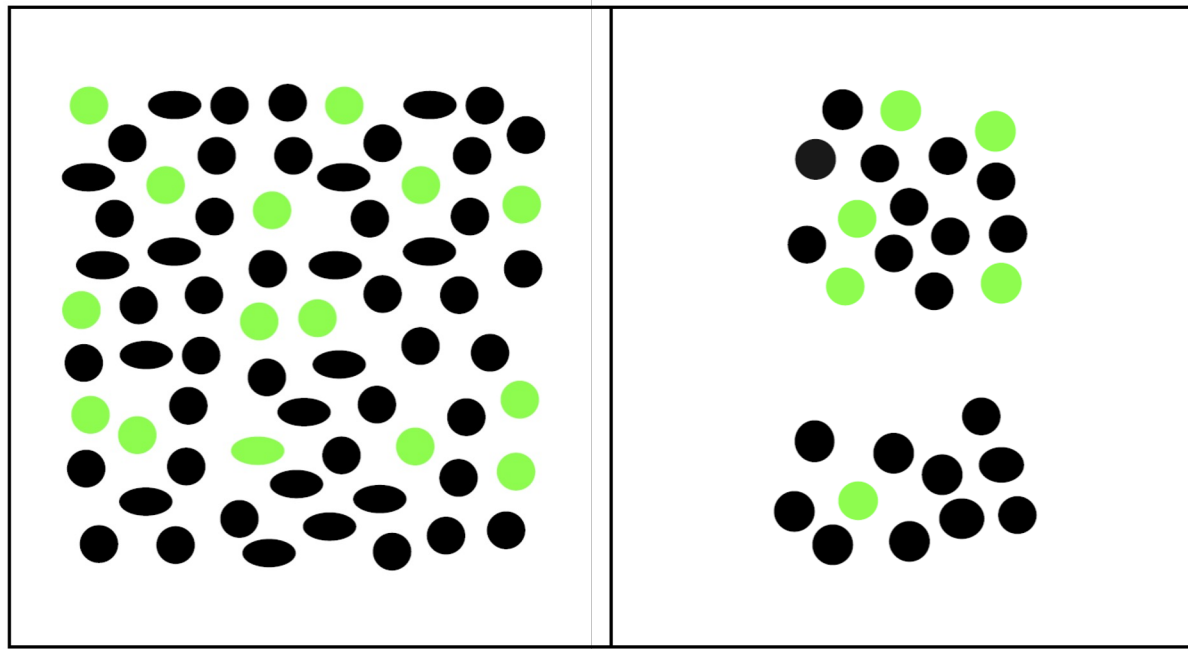


[Colin 2008]

Find the green squares

Effective Use of Conjunctions

- Not all conjunctions are as effective as the others.



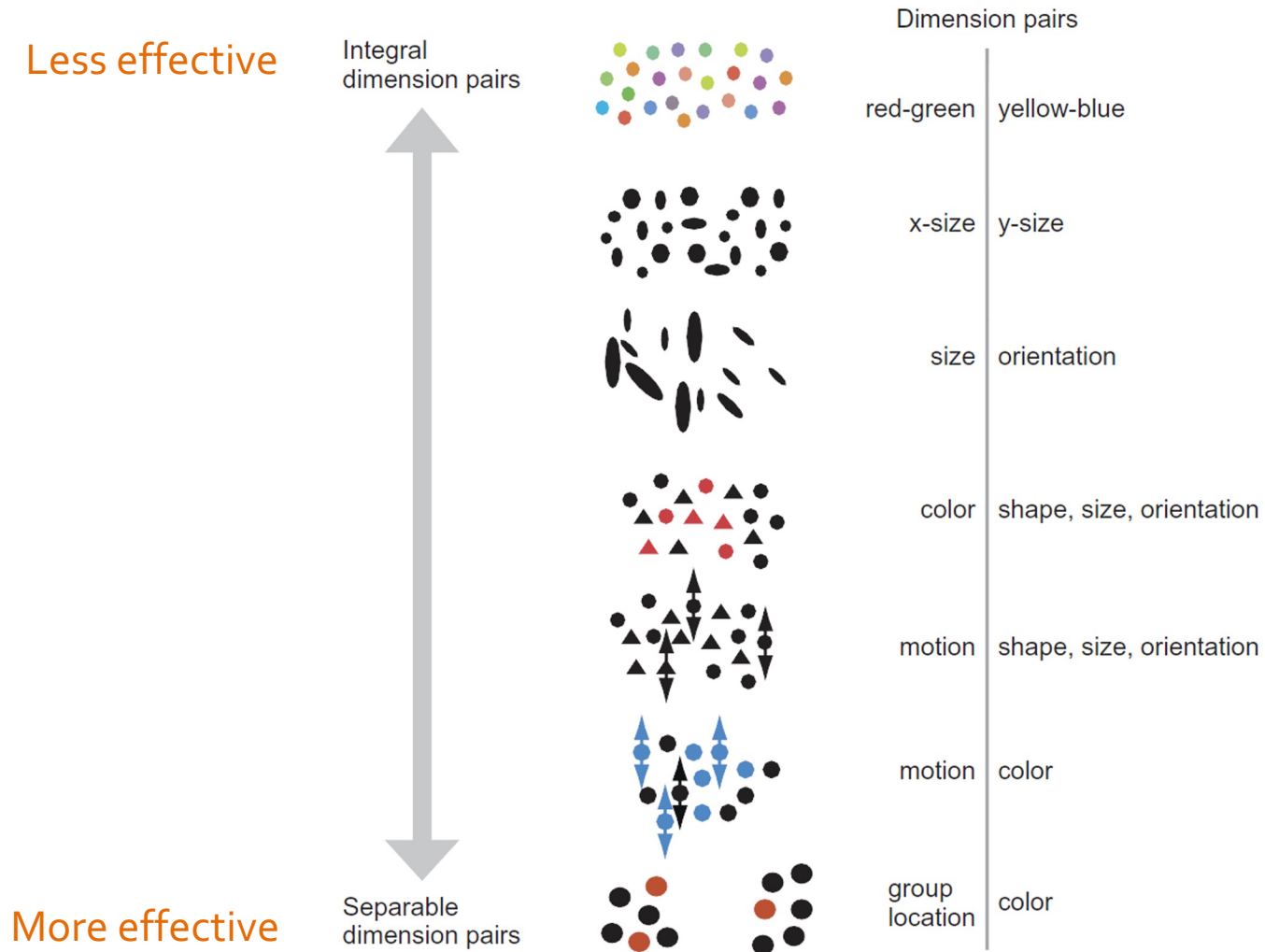
Find the green ellipse

Color + Shape

Find the green target in the lower cluster

Color + Group location

Effective Use of Conjunctions



[Colin 2013]

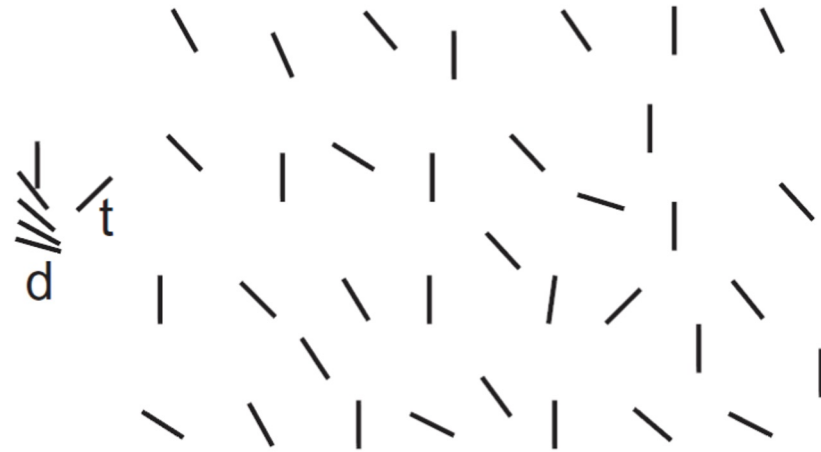
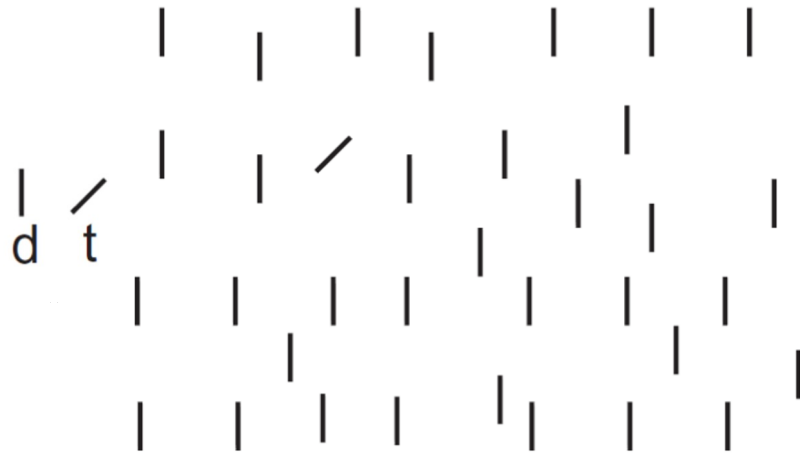
Preattentive Salience

- Stronger effects:
 - Color, orientation, size, contrast, motion/blinking

Use strong preattentive features/cues before
weak ones to facilitate ease of search

- But depends also on degrees of differences between the target and the others
 - E.g. large vs. small color differences

Preattentive Saliency



[Colin 2008]

t: target

d: distractor

Preattentive Salience

- Other factors determining if a feature stands out preattentively:
 - Degree of difference of the target from the nontargets (distractors)
 - Degree of difference of the nontargets from each other

Preattentive Salience

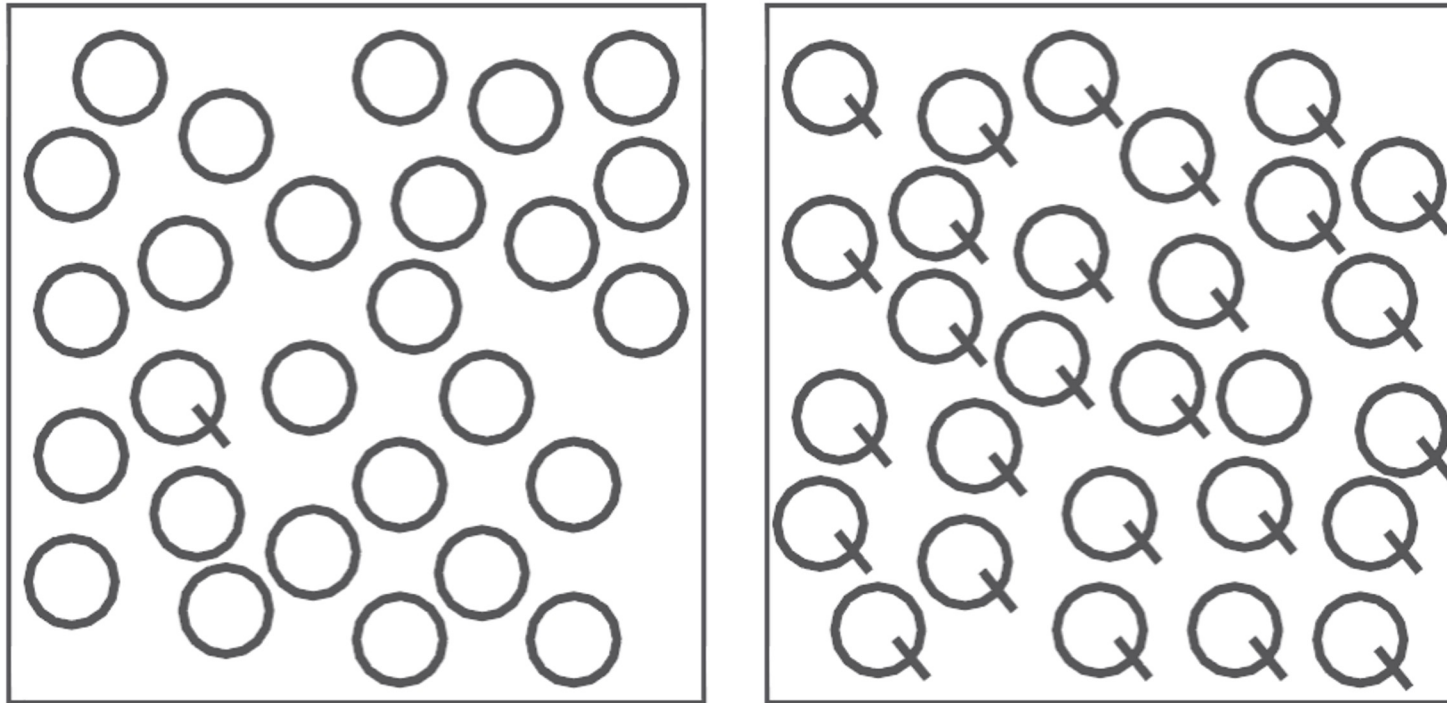
Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Pellentesque ornare nisi erat, eget luctus magna faucibus eu.
Aliquam auctor mollis nulla, at pretium turpis. Sed porttitor
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Nam dapibus metus lectus, vel fringilla turpis facilisis in.
Phasellus non velit dolor.

Preattentive Salience

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Pellentesque ornare nisi erat, eget luctus magna faucibus eu.
Aliquam auctor mollis nulla, at pretium turpis. Sed porttitor
nec justo quis mollis. Lorem ipsum dolor sit amet, consectetur
adipiscing elit. Suspendisse mollis vulputate justo sed aliquet.
Proin a ante nec tellus condimentum lobortis. In ornare sit
amet lorem et sollicitudin. In mollis risus at risus cursus, sed
laoreet magna molestie. Sed gravida metus a gravida aliquet.
Nam dapibus metus lectus, vel fringilla turpis facilisis in.
Phasellus non velit dolor.

Asymmetric Preattentive Cues

- The “reverse” of a preattentive feature is not necessary preattentive.



Asymmetric Preattentive Cues

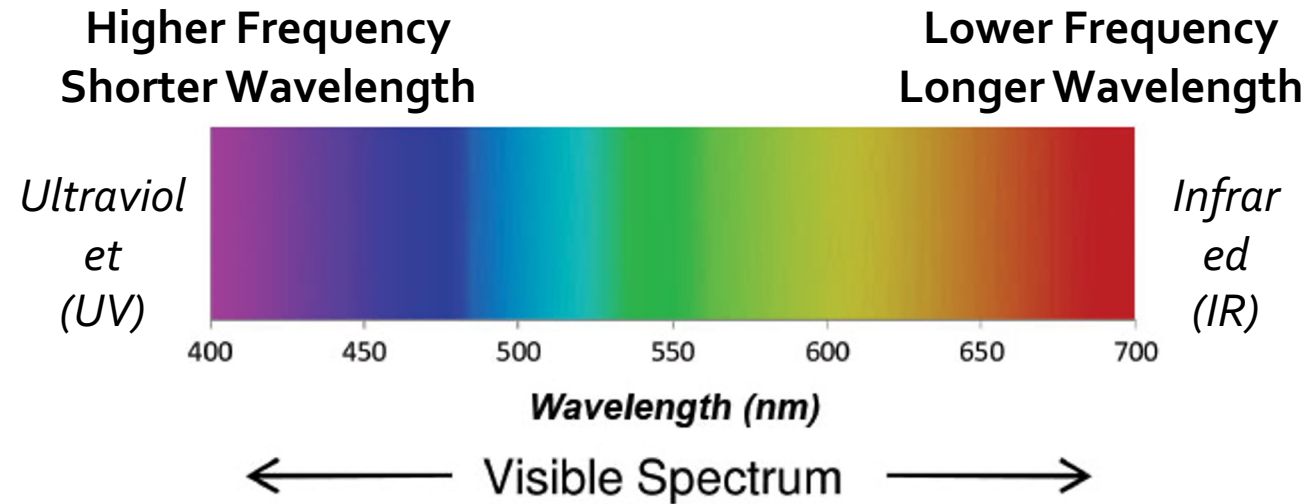
- Examples
 - Underlining words in a paragraph
 - Big targets surrounded by small ones
- Use **positively** asymmetric preattentive cues for highlighting

Color



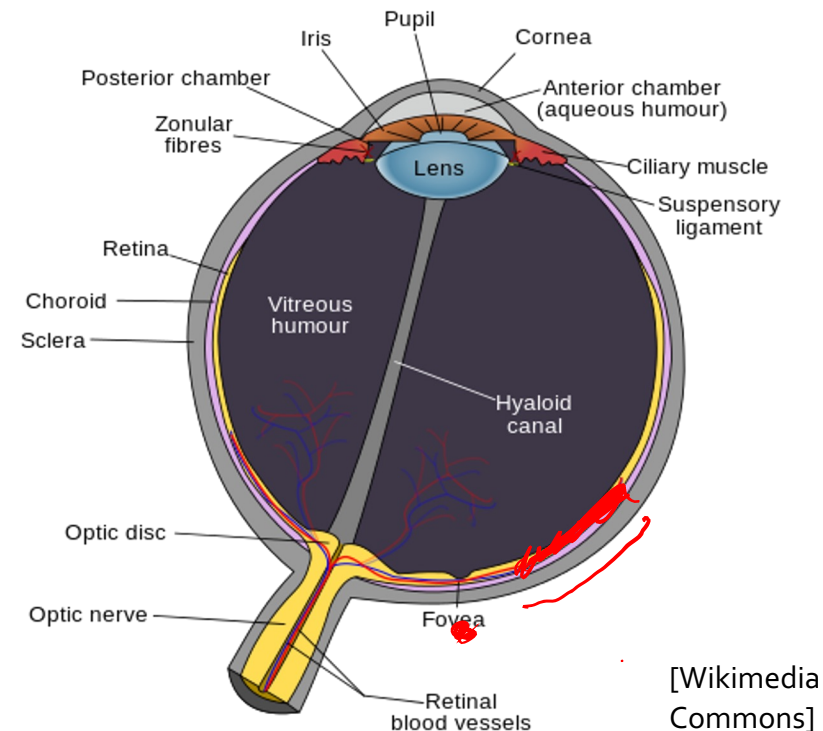
Visible Light Spectrum

- A spectrum of wavelengths from 370 to 730 nanometers (nm)



The Visual System

- How do we see?
 - Light travels through cornea, pupil, lens, retina, optical nerves, then brain
 - Imaging sensors on the retina: rods & cones



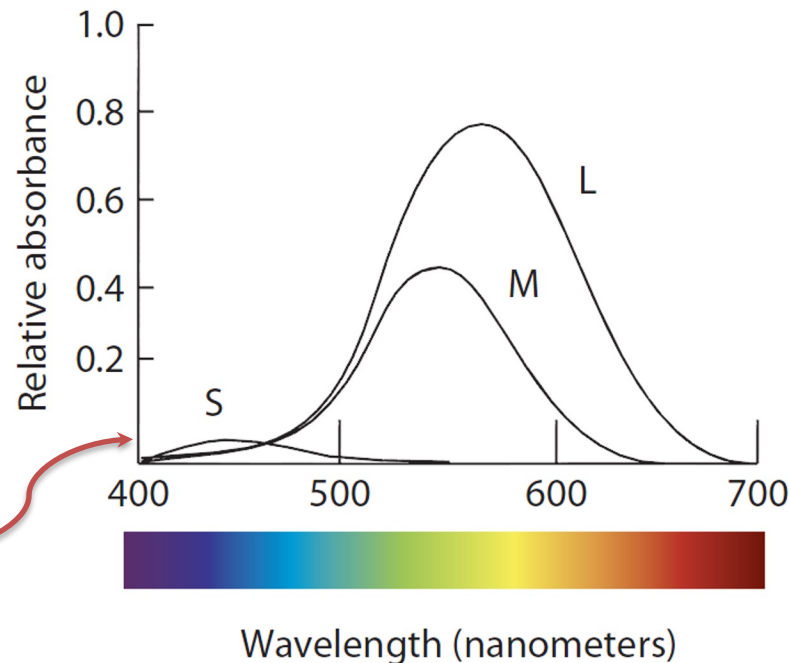
Rods and Cones

- **Cones**
 - active at normal light levels
 - color vision involves cone only
- **Rods**
 - sensitive at low light levels
 - not sensitive to color
 - responsible for our dark-adapted vision
 - low influence on color perception
- There is an uneven distribution of cones and rods in the retina

Cone Response

- Three classes of cone receptors: long (L), medium (M), and short (S) wavelength-sensitive cones.

Sensitivity functions of the three cone receptors

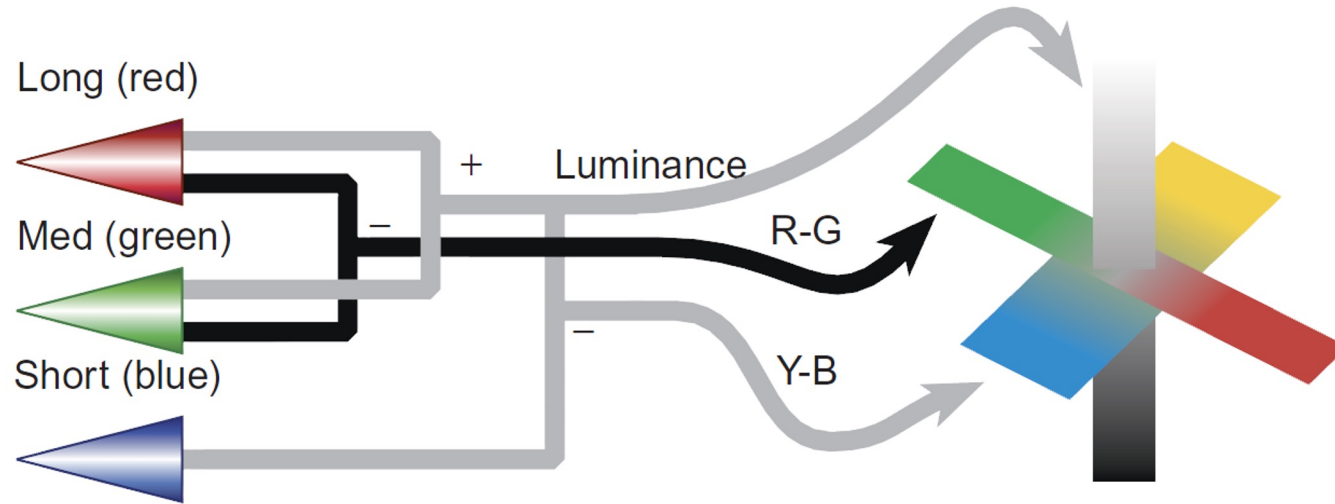


This is why we should not show blue over a dark background since S cone receptors are much less sensitive.

[Ware, 2008]

Opponent Color Theory

- Input from the cones is processed into 3 distinct channels: **luminance** (black-white), **red-green**, and **yellow-blue**.
- Explain how we perceive some of the color effects, e.g., brightness contrasts or chromatic contrasts.

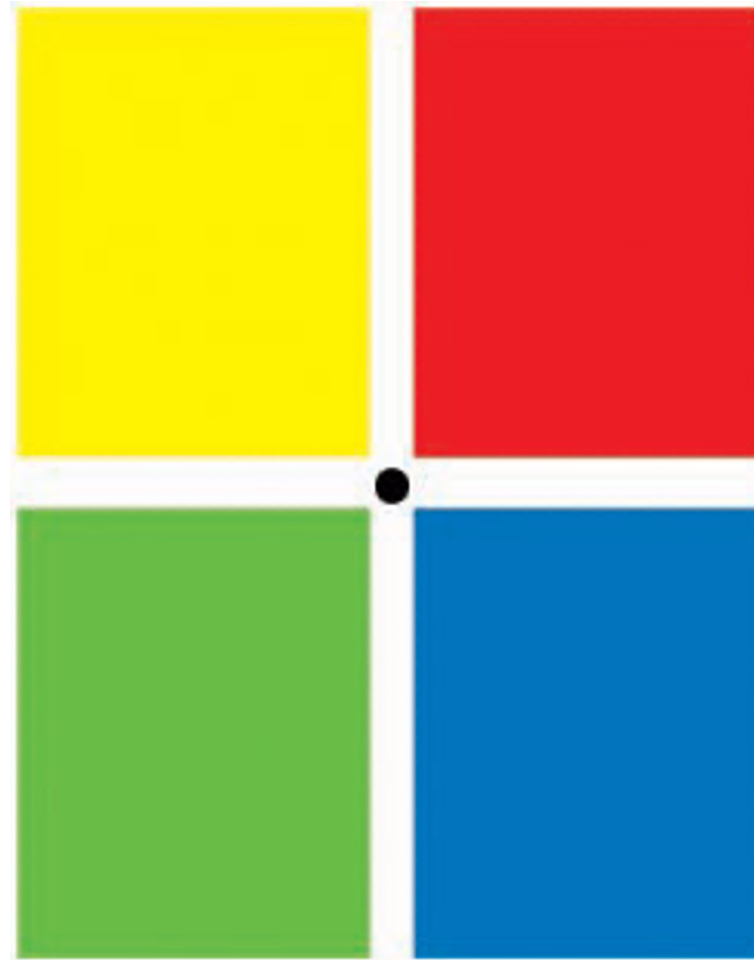


[Ware,
2013]

Opponent Color Theory

- An experiment

The afterimage caused by saturating the red-green, blue-yellow, and luminance channels.

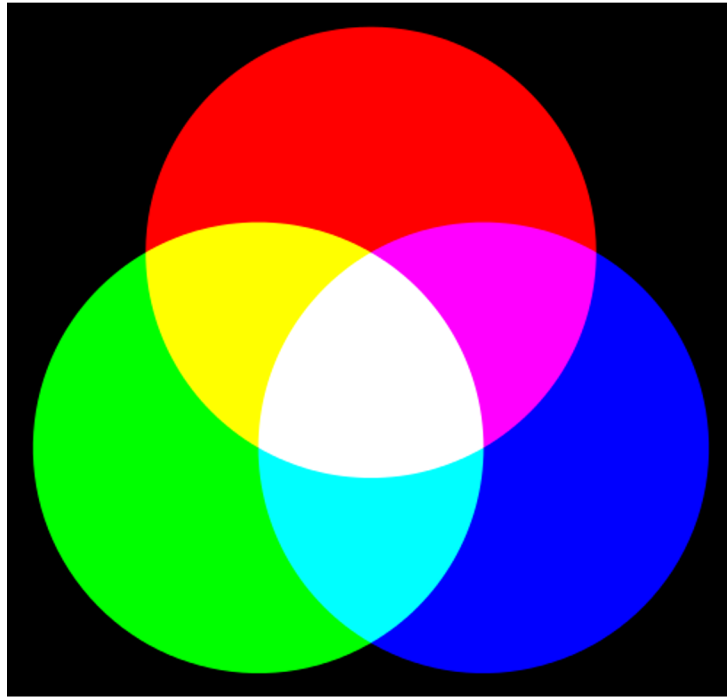


[Stone]

Primary Colors

- Red, Green, Blue
 - liquid crystal, CRT displays
- Cyan, Magenta, Yellow
 - color printing
- Red, Yellow, Blue
 - paint
- Orange, Green, Violet
 - color photography

Physical Color Spaces



Additive, RGB

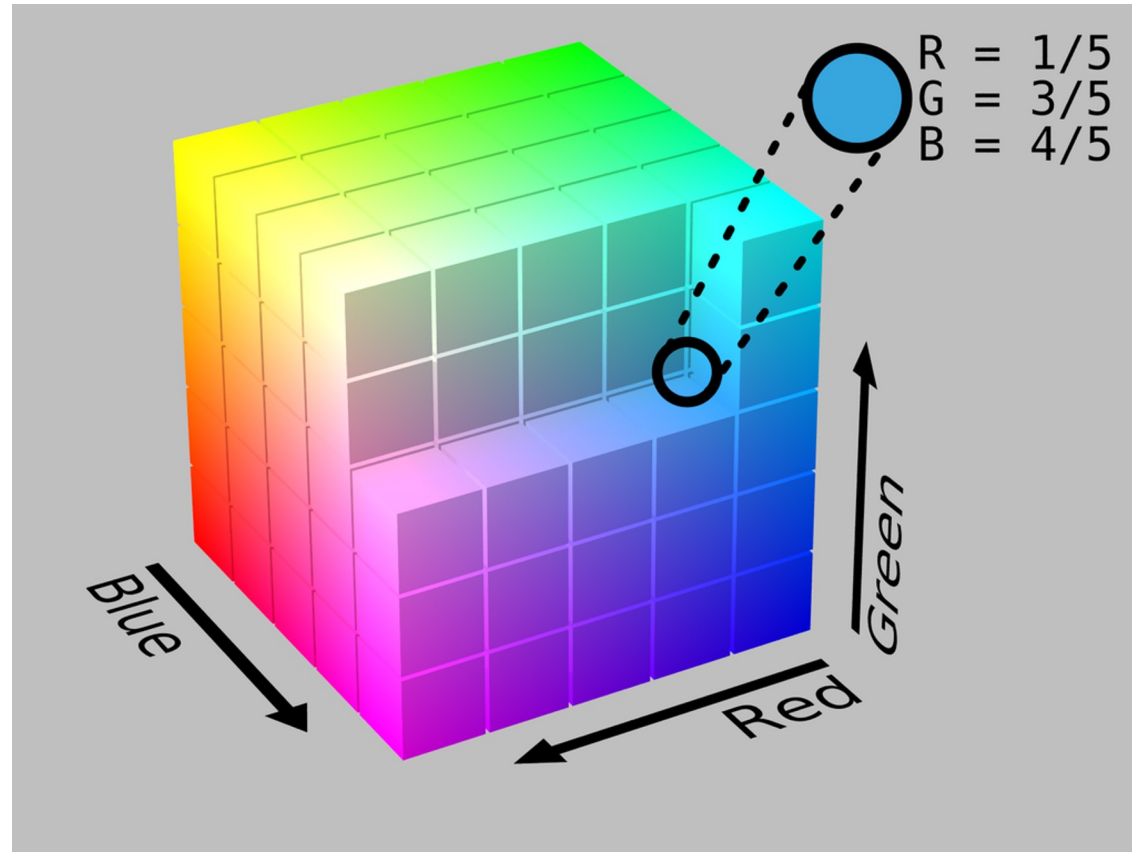
For media that emits light



Subtractive, CMY

For media that reflects light

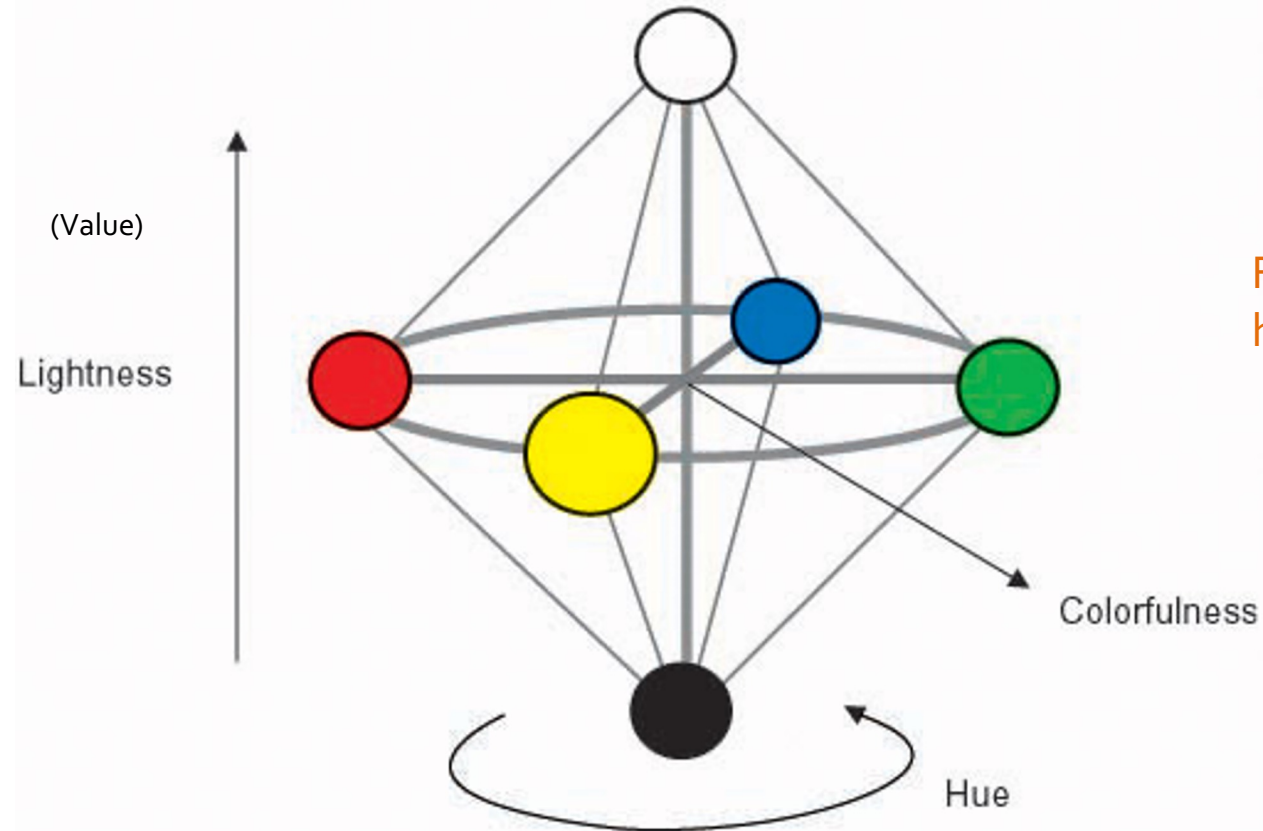
RGB Color Space



Not intuitive to describe a color

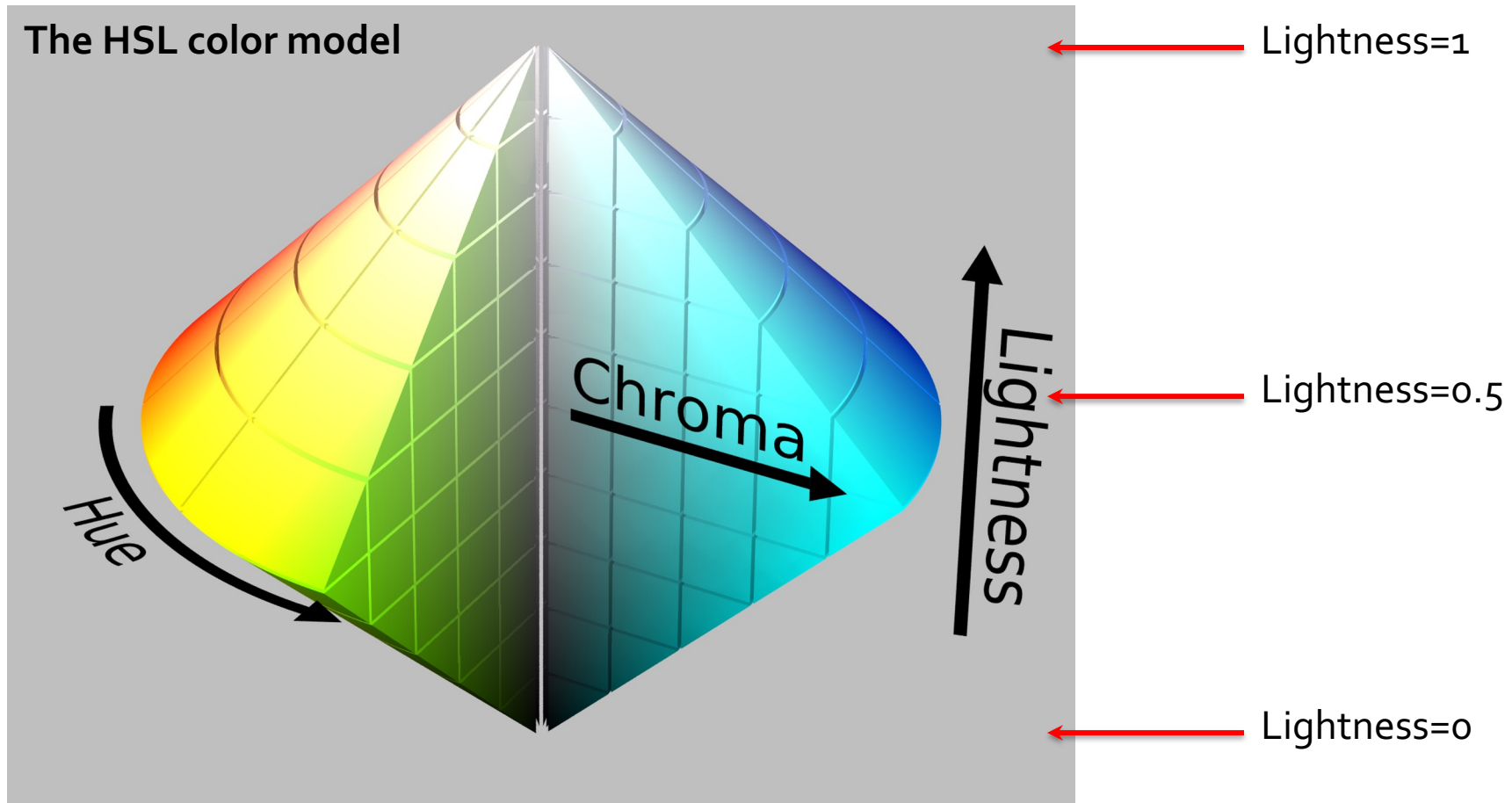
Perceptual Color Spaces

- How human perceive a color:
hue, lightness and **colorfulness (color saturation)**



[Stone]

Perceptual Color Spaces

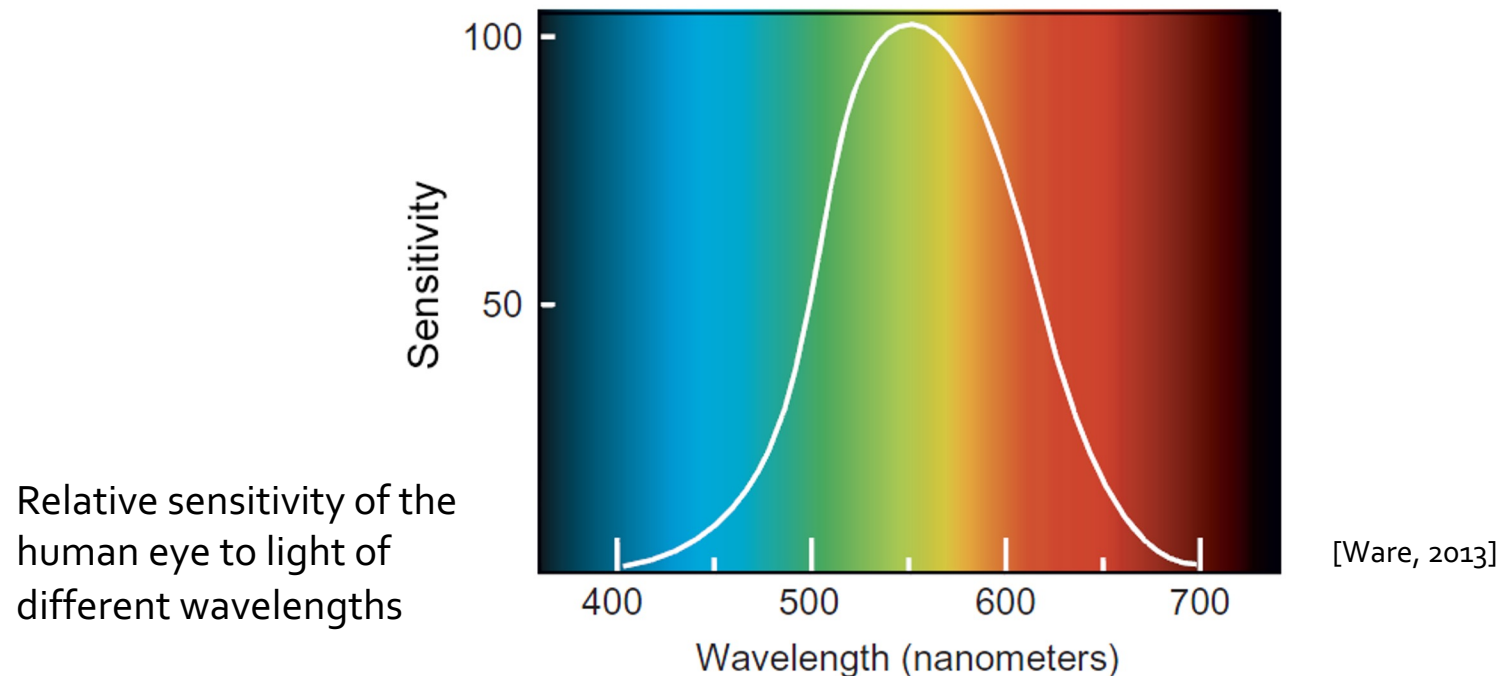


SharkDderivative work

https://en.wikipedia.org/wiki/HSL_and_HSV#/media/File:HSL_color_solid_dblcone_chroma_gray.png

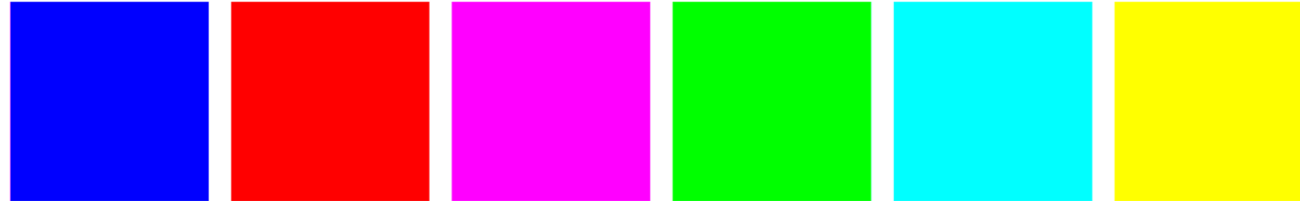
Lightness Scales

- **Lightness / Brightness** – (qualitative) **perceived** reflectance of a surface
- **Luminance** – (quantitative) **measured** amount of light energy weighted by the spectral sensitivity function of the human eye



Lightness vs. Luminance

Colors of different hues



Same lightness, e.g., 0.5

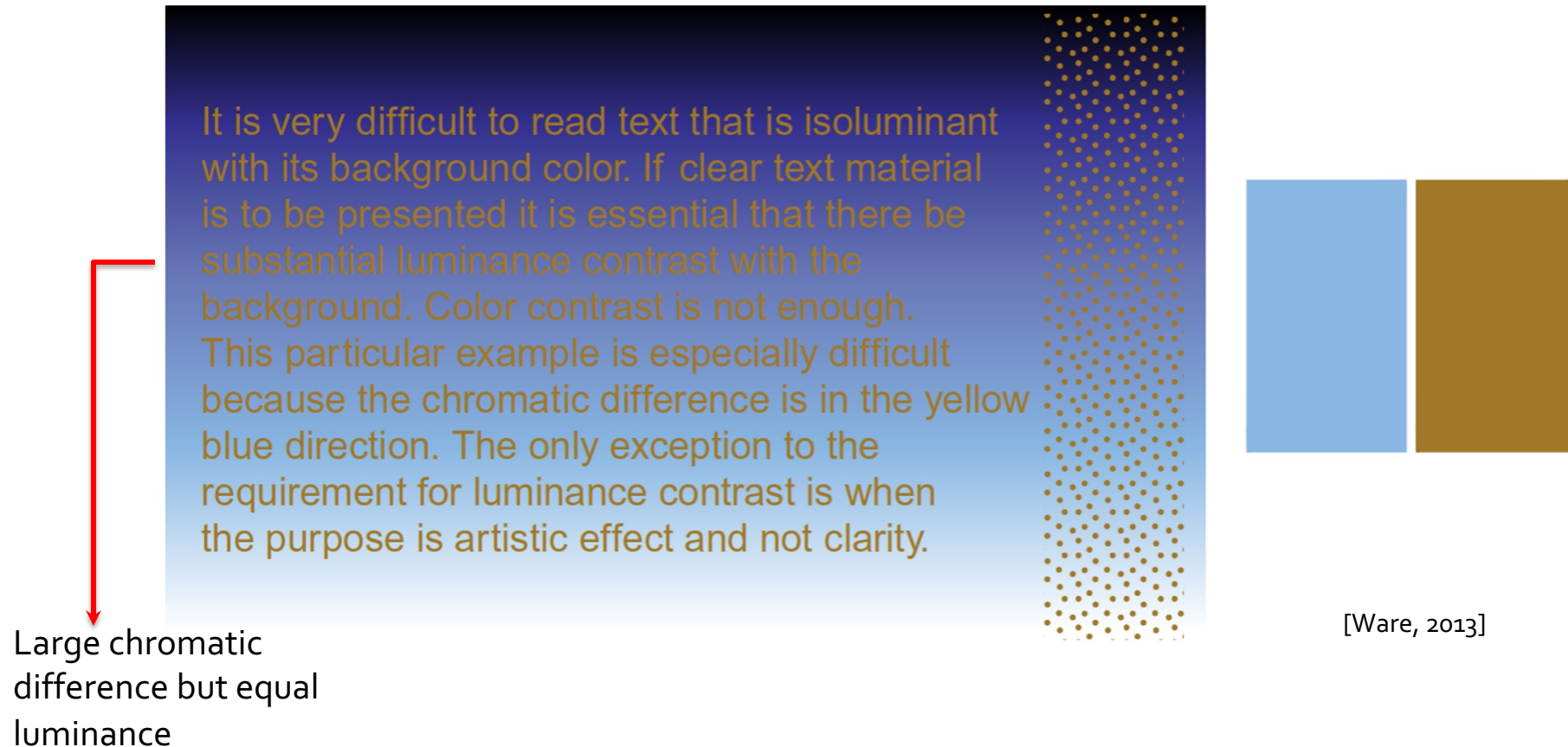


Different luminance



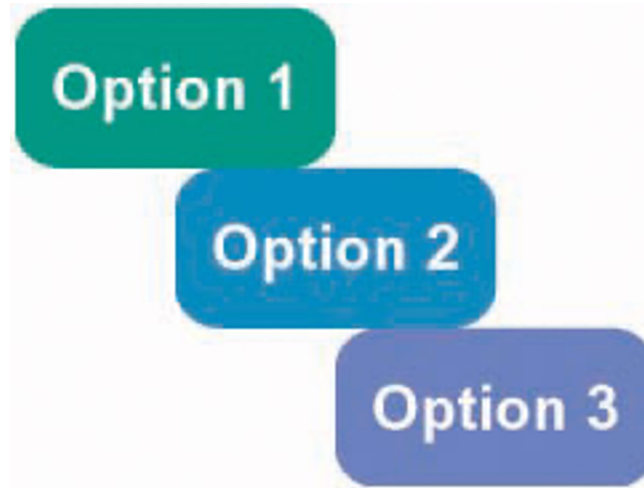
Contrast

- Difference in luminance
- Determines legibility and sharpness



Equal Luminance Value

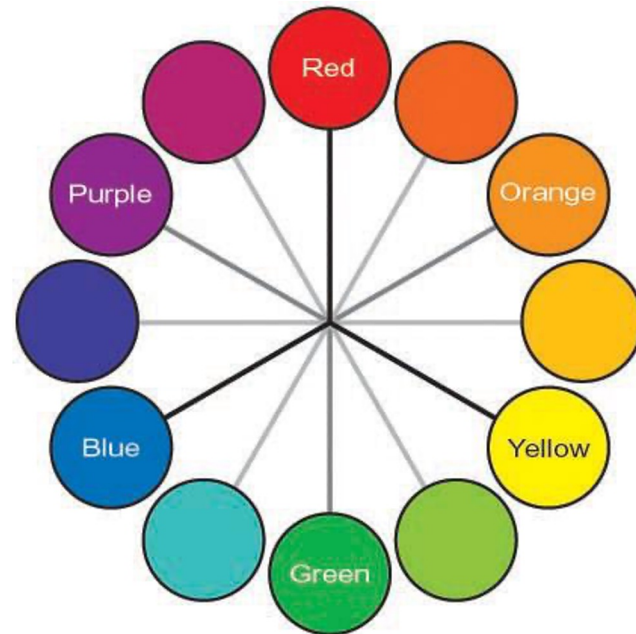
- Gives same importance or emphasis



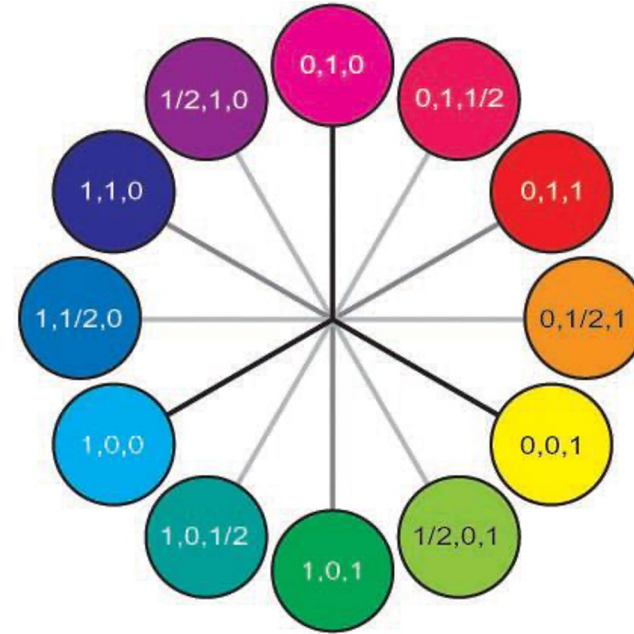
[Stone]

Color Wheel

- Analogous colors lie close and contrasting colors lie opposite.



Primary colors: R, G, B

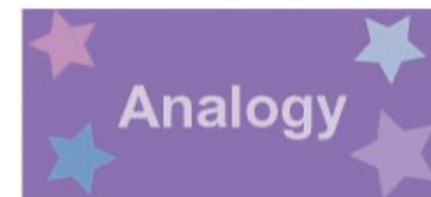
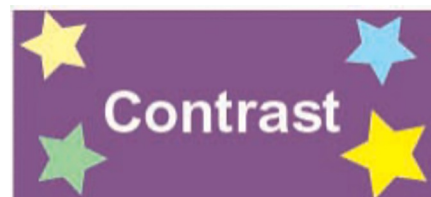


Primary colors: C, M, Y

[Stone]

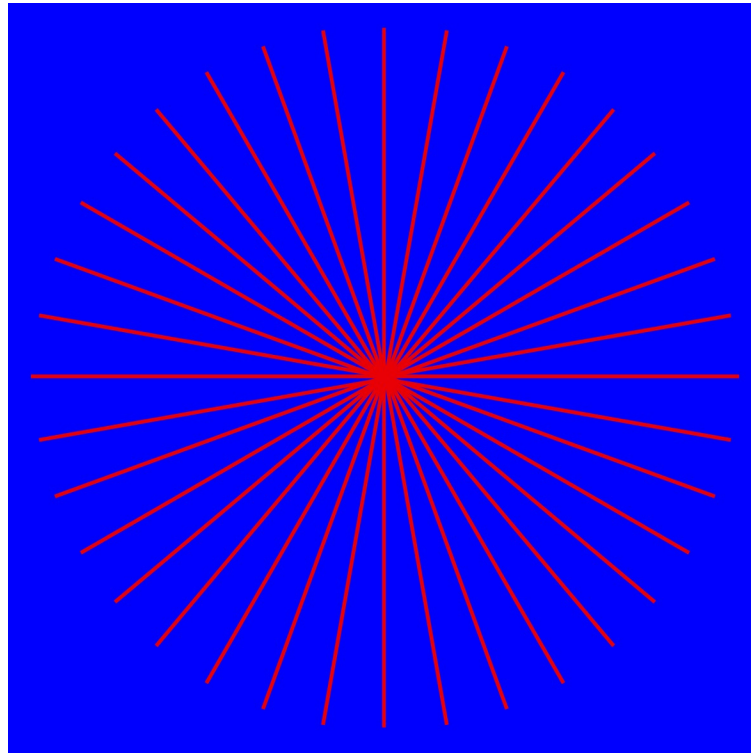
- Color Harmony

- **Contrasts:** focus attention
- **Analogy:** unify design



Color on Color

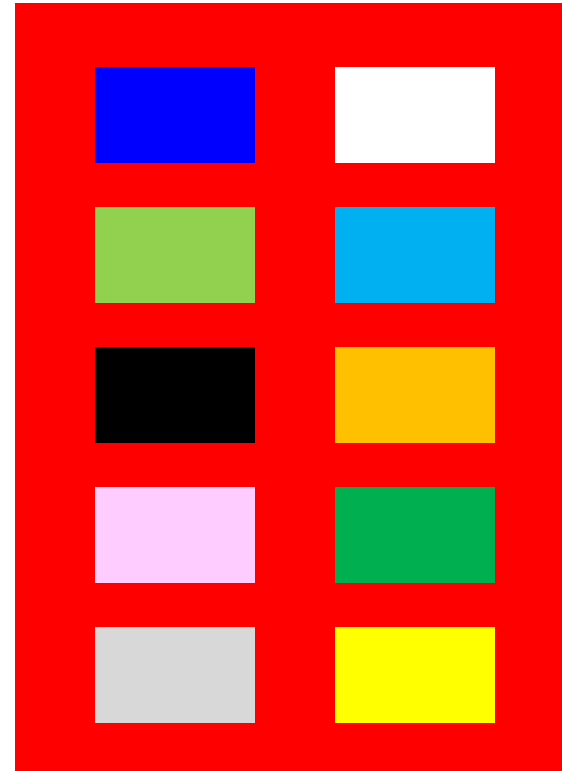
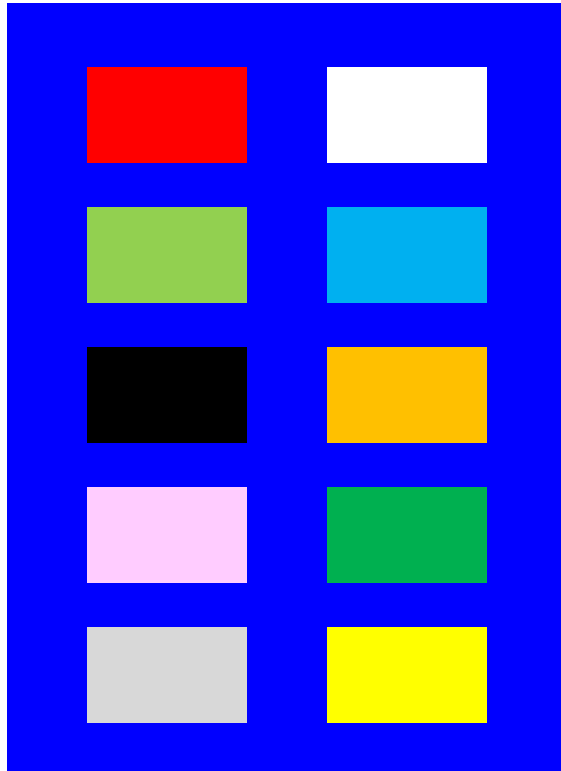
- **Chromatic aberration** – different colors focus at a different depth in the eye



Difficult to focus on both red lines
and blue background

Color on Color

- Chromatic aberration



Color on Color

- Blue-sensitive cones are few and weak.

**Lorem ipsum dolor sit amet,
consectetur adipiscing elit.
Pellentesque ornare nisi erat,
eget luctus magna faucibus eu.**

**Lorem ipsum dolor sit amet,
consectetur adipiscing elit.
Pellentesque ornare nisi erat,
eget luctus magna faucibus eu.**

Color on Color

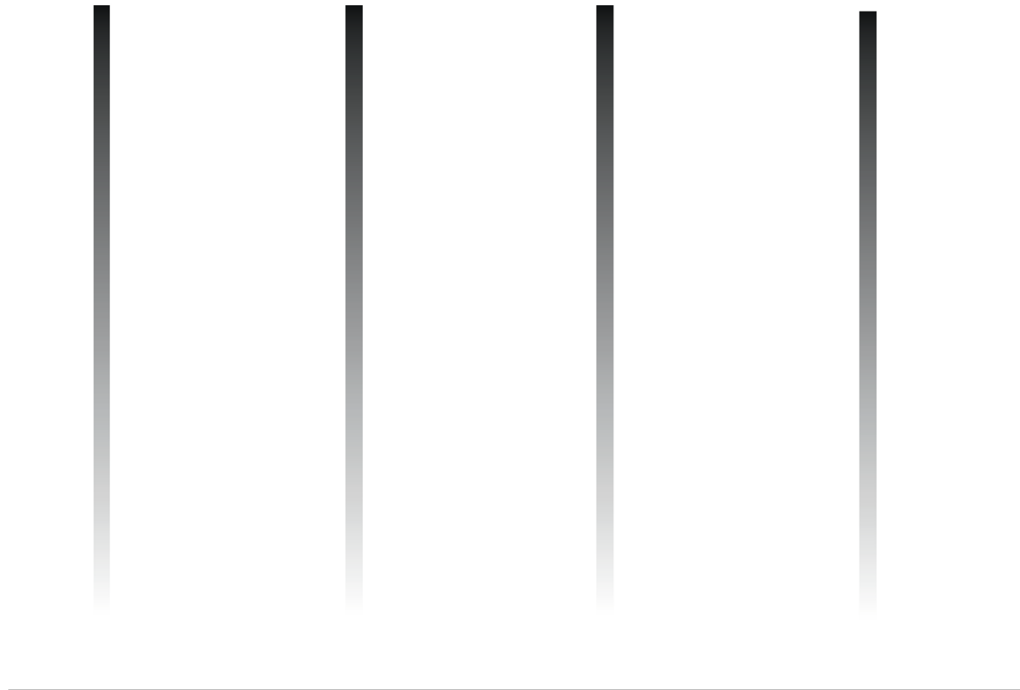
- Yellow is the lightest of all pure hues.

**Lorem ipsum dolor sit amet,
consectetur adipiscing elit.
Pellentesque ornare nisi erat,
eget luctus magna faucibus eu.**

**Lorem ipsum dolor sit amet,
consectetur adipiscing elit.
Pellentesque ornare nisi erat,
eget luctus magna faucibus eu.**

Color on Color

- Perceived differences are enhanced where the values are close to the background gray value.



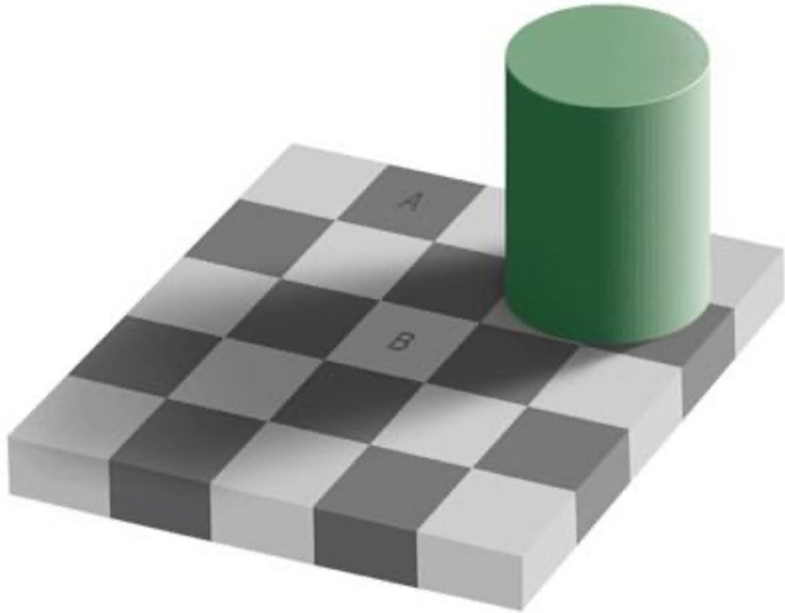
Color on Color

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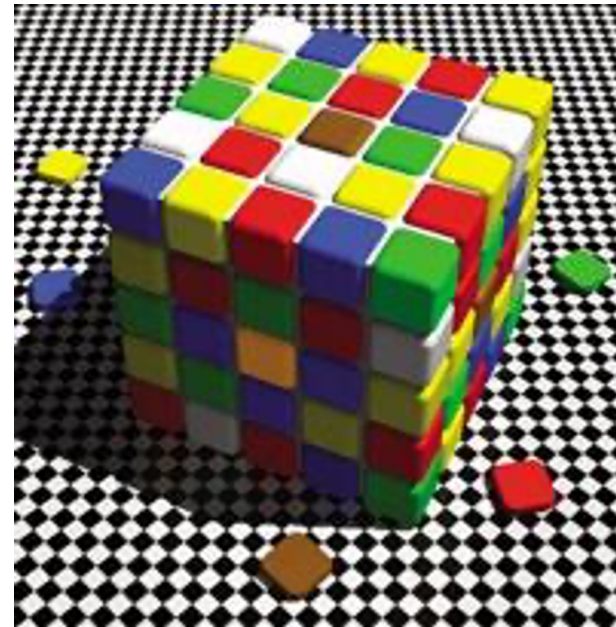


Color on Color

- Contextual effects



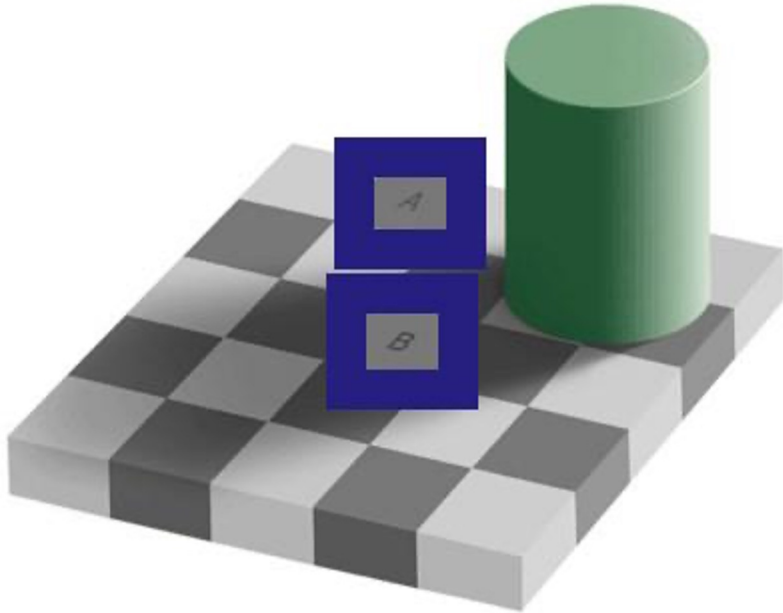
[E.A. Adelson,
MIT]



[R. B. Lotto,
www.lottolab.org]

Color on Color

- Contextual effects



[E.A. Adelson,
MIT]



[R. B. Lotto,
www.lottolab.org]

Color on Color

- How to reduce unpredictable effect of color on color?
 - Avoid doing color on color entirely
 - Layering colors only with neutral colors
- Ensure that edges are visible and text is legible

Purpose of Color

- Purpose of color in information design:
 - To label (color as noun)
 - To measure (color as quantity)
 - To represent or imitate reality (color as representation)
 - To enliven or decorate (color as beauty)
- Most important of all is that color should do no harm

[Tufte, Envisioning Information]

To Label

- For categorical data
- A preattentive process, we can group data by colors at a glance



[Stone]

To Label

- **Stroop effect:** your right brain is trying to say the color, while your left brain insists on saying the word

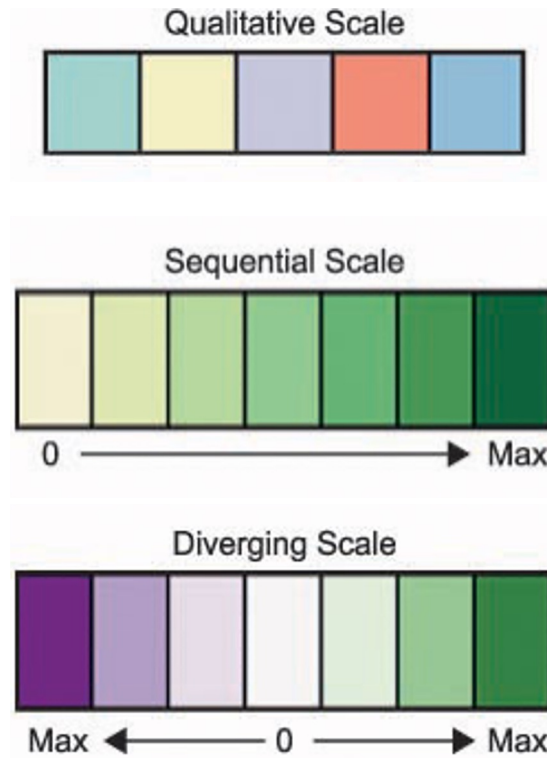
YELLOW BLUE ORANGE
BLACK RED GREEN
PURPLE YELLOW RED
ORANGE GREEN BLACK
BLUE RED PURPLE
GREEN BLUE ORANGE

Name the color of each word

- Stronger pre-attentive cues by text than by color
- Avoid color labeling that conflicts with convention

To Quantify

- Color scales: sequences of color values to indicate relative quantity



For labeling

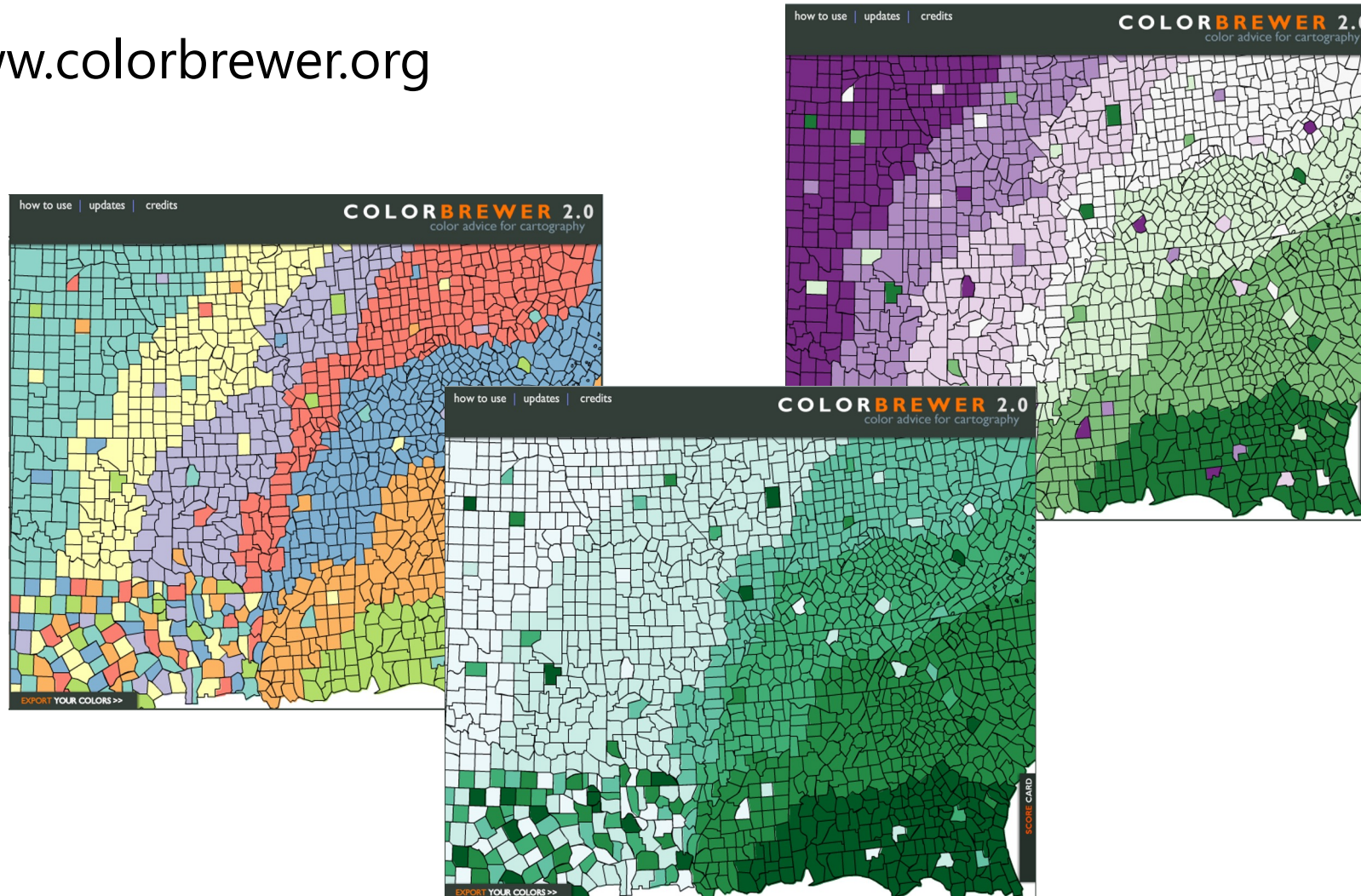
For indicating quantity

Two sequential scales cross-fade through a neutral color

[Stone]

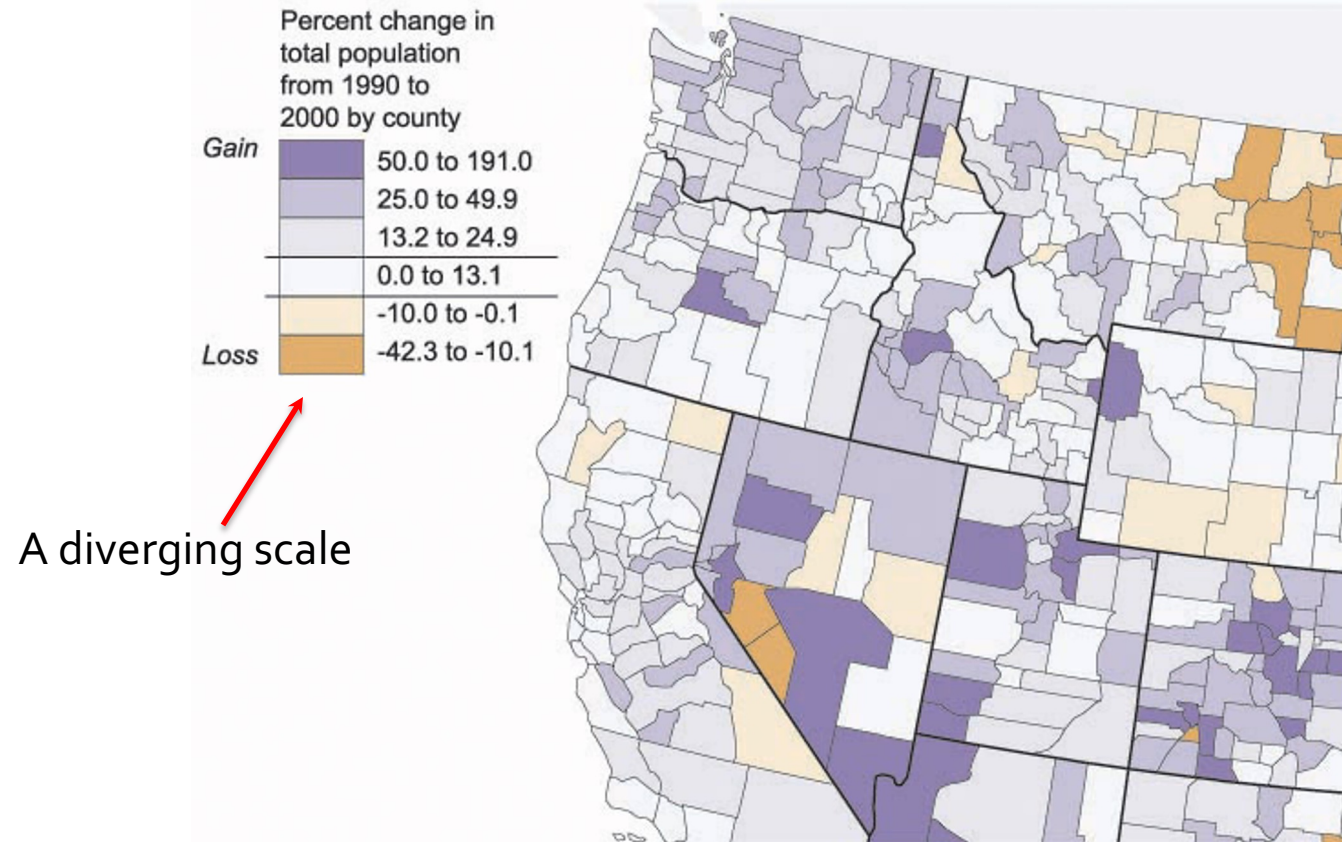
To Quantify

- <http://www.colorbrewer.org>



To Quantify

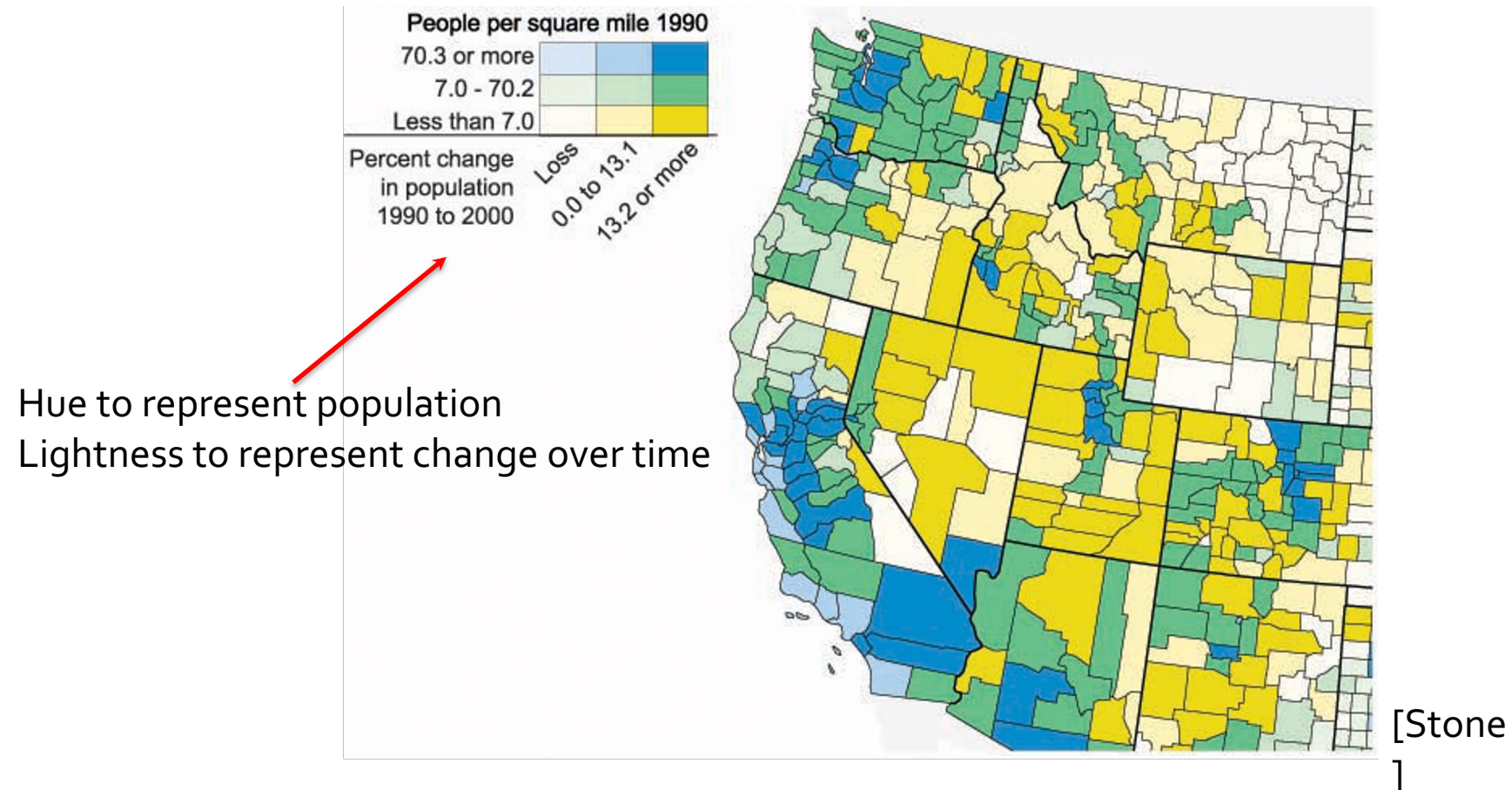
- Univariate data



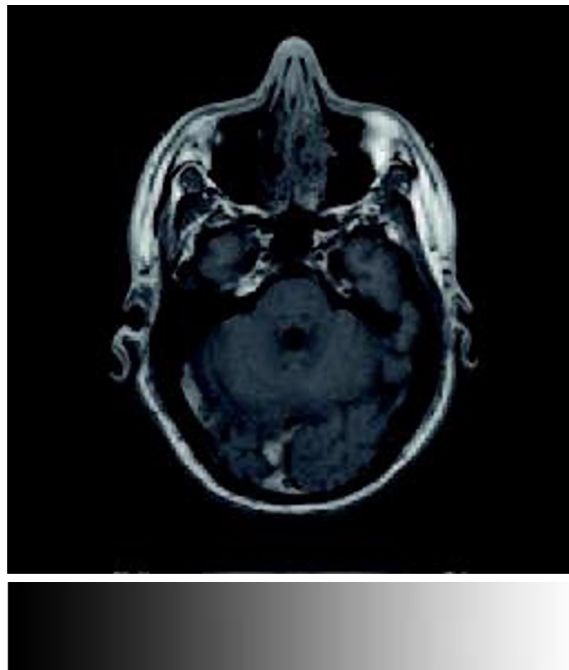
[Stone
]

To Quantify

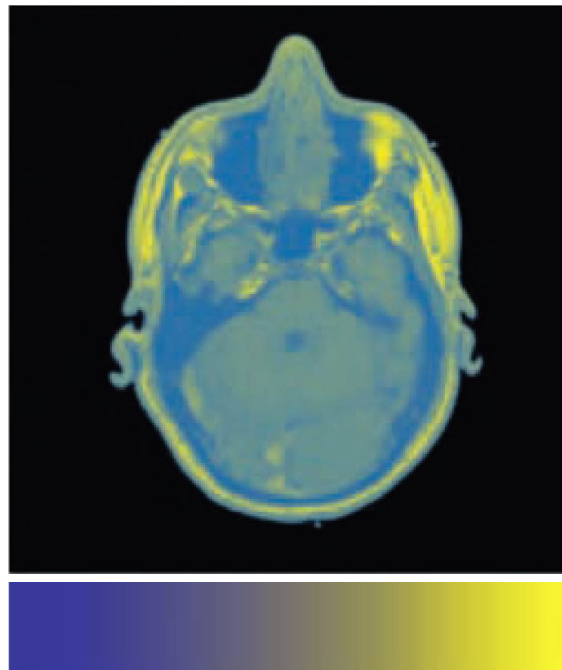
- Multivariate data



Bad Use of Color

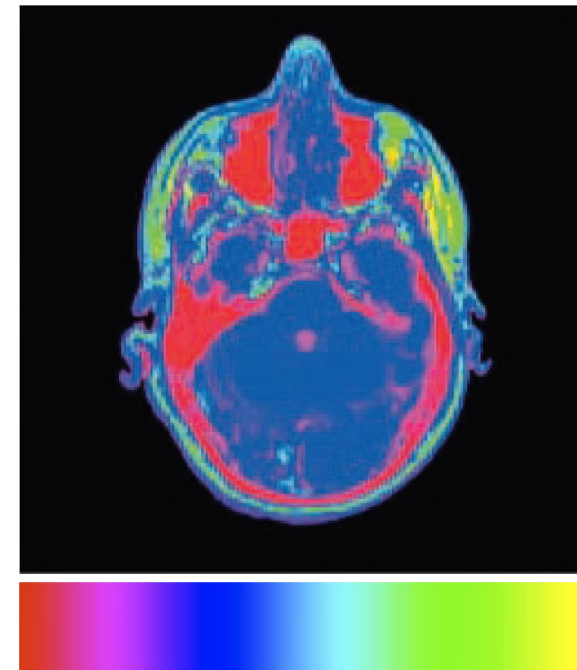


Original MRI data



Pseudocolor
blue-yellow gradient

Similar lightness scale



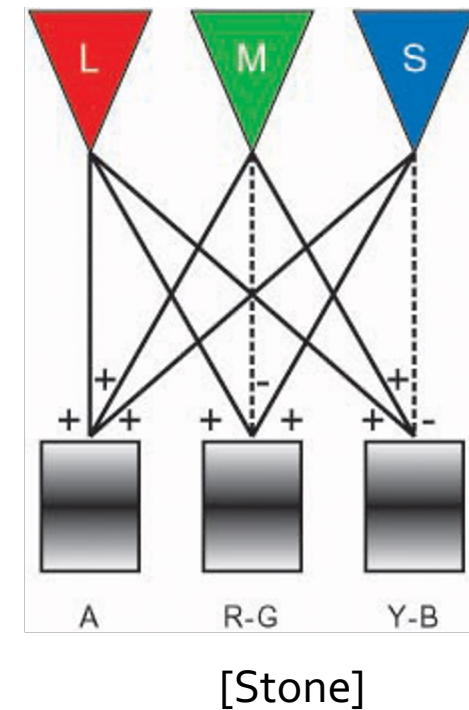
Pseudocolor
rainbow color

Obscures the original data

[Stone]

Color Blindness

- About 10% of people (mostly men) has color vision deficiencies.
- Most have problems in processing red-green variations.
- Most have no problem in processing luminance information.
- **Rule 1:** maintain **sufficient value contrast** when reduce to shades of gray
- **Rule 2:** reinforce color encoding with **position, size, shape**



References

- Matthew Ward, Georges Grinstein and Daniel Keim, "*Interactive Data Visualization: Foundations, Techniques, and Applications*", 2010 [Chapter 3]
- Colin Ware, "*Visual thinking for design*", 2008.
- Colin Ware, "*Information Visualization – Perception for Design*", 2013.
- Maureen C. Stone, "*A Field Guide to Digital Color*", 2003.
- Edward R. Tufte, "*Envisioning Information*", 1992.